Spatial one-hot nucleotide representations of DNA enable robust and interpretable taxonomic classification with 1D convolutional networks

Matheus de Lima[1], Ulisses Nunes da Rocha[2], Márcio Dorn[1,3*]

[1*]Structural Bioinformatics and Computational Biology Lab, Institute of Informatics, Universidade Federal do Rio Grande do Sul (UFRGS), Porto Alegre, Rio Grande do Sul, Brazil.

[2]Department of Computational Biology and Chemistry, Helmholtz Centre for Environmental Research (UFZ), Permoserstr. 15, Leipzig, 04318, Germany.

[3]Center for Biotechnology, Universidade Federal do Rio Grande do Sul (UFRGS), Porto Alegre, Rio Grande do Sul, Brazil.

*Corresponding author(s). E-mail(s): mdorn@inf.ufrgs.br;
Contributing authors: matheuslimaaa13@gmail.com;
ulisses.rocha@ufz.de;

Abstract

High-cardinality viral taxonomic classification requires models capable of distinguishing closely related genomic sequences while maintaining predictive performance as sequence quality deteriorates. We investigated base-level one-hot DNA representations combined with 1D adaptations of ResNet50 and ConvNeXt in a closed-set taxonomic classification setting comprising 13,683 viral species and 2,702 genera. Models were evaluated on held-out 512-bp sequence fragments. We further assessed robustness under controlled nucleotide resampling, the effect of noise-aware fine-tuning, the organization of latent representations, the predictive contribution of first-layer sequence patterns through matched perturbation, and sliding-window aggregation for variable-length genomic sequences.

On uncorrupted species-level fragments, the evaluated ResNet50 model achieved the highest accuracy (84.88%). Its performance, however, decreased more rapidly under nucleotide resampling than that of the ConvNeXt configurations. At $\epsilon = 0.15$, corresponding to an expected effective nucleotide-change rate of 11.25%, noise-aware ConvNeXt retained 17.38% accuracy, compared with 3.87%

for noise-aware ResNet50 and 0.30% for standard ResNet50. Analysis of the pre-classifier representations revealed weak global class separation but substantial local taxonomic consistency, with higher k-nearest-neighbor (KNN) accuracy for the ConvNeXt configurations. Matched perturbation further identified first-layer filter-derived sequence patterns whose disruption reduced true-class probability more strongly than perturbation of matched random regions. Sliding-window aggregation extended the fixed-length classifiers to variable-length genomic sequences, with standard ResNet50 reaching 72.69% species-level mean-logit accuracy on the evaluated long-sequence [**Nota v2:** [Coerência] Destacar este valor no resumo é arriscado enquanto não se esclarecer se as sequências longas coincidem com os genomas do treino e por que os modelos ConvNeXt caem para 3,5–20% nesse conjunto. Considerar reformular a frase como demonstração de viabilidade, sem ênfase no valor numérico.] dataset.
The evaluated configurations exhibited a pronounced trade-off between clean-sequence accuracy and robustness to controlled nucleotide corruption. These results support base-level one-hot 1D convolutional models as a useful framework for studying high-cardinality closed-set viral taxonomic classification, robustness, and learned sequence representations.

Keywords: Viral Taxonomic Classification; Deep Learning; 1D Convolutional Neural Networks; Genomic Sequence Representation; Sequence Corruption; Robustness; Latent Representations; Sequence Pattern Perturbation

[**Nota v2:** [Contribuição] Avaliação geral: a contribuição do trabalho está na *avaliação*, e não no método. A representação *one-hot* e as CNNs para classificação taxonômica não são novas; o que é novo é (i) a escala (13,683 espécies virais), (ii) o *trade-off* documentado entre acurácia em dados limpos, robustez à corrupção e comportamento da confiança, e (iii) o protocolo de avaliação (corrupção com taxa efetiva corrigida, perturbação pareada, análise latente local e global). Recomenda-se reenquadrar título, resumo, Introdução e Conclusão em torno dessa mensagem, por exemplo como "avaliação sistemática de robustez e confiança de CNNs em classificação viral de alta cardinalidade". Três acréscimos tornariam a contribuição sólida diante dos revisores: uma *baseline* clássica sob o mesmo protocolo de ruído (Kraken2), uma partição por genoma de origem e intervalos de confiança.]

[**Nota v2:** Pendências editoriais gerais. (i) Confirmar se o Ulisses deseja incluir o grupo (Microbial Systems Data Science) na afiliação. (ii) Periódicos BMC, incluindo a *BioData Mining*, exigem resumo estruturado (*Background*, *Results*, *Conclusions*); os três parágrafos do resumo já seguem essa lógica e podem receber os subtítulos. Na v2, a seção inicial foi renomeada para *Background* e os Métodos foram movidos para antes dos Resultados. (iii) Sugere-se moderar o título: "enable robust" não é sustentado pelos resultados (acurácia de 3,87–17,38% em $\epsilon = 0{,}15$ e o próprio resumo descreve um *trade-off*). Exemplo: "Base-level one-hot 1D convolutional networks for high-cardinality viral taxonomic classification: accuracy, robustness, and learned representations". (iv) Todos os resultados derivam de uma única execução de treinamento por configuração; recomenda-se reportar intervalos de confiança (p.ex., *bootstrap* sobre o conjunto de teste) e, idealmente, variabilidade entre sementes.]

1 Background

Advances in high-throughput DNA sequencing have enabled genomic data to be generated at increasingly large scales and across a broad range of short- and long-read

technologies [1, 2]. In metagenomic analysis, converting these sequence reads into reliable taxonomic assignments remains a major computational challenge, particularly as reference databases increase in size and taxonomic resolution [3, 4].

Widely used alignment-free methods address this problem by matching sequences against reference databases. Kraken 2, for example, uses exact k-mer matches together with lowest-common-ancestor assignment to achieve efficient taxonomic classification [5]. Exact nucleotide matching, however, can be disrupted by sequence errors or other local perturbations. This issue is particularly relevant for long-read sequencing, where substitutions, insertions, deletions, and context-dependent errors may affect sequence comparison and downstream classification [6–8]. [**Nota v2:** [Ref.] As três referências sustentam a existência e o perfil dos erros de *long reads* (Delahaye e Nicolas descrevem erros e vieses; NanoSim e Badread são simuladores), mas nenhuma avalia o efeito desses erros sobre a *classificação taxonômica*. Para essa parte da frase, e para a afirmação anterior de que erros prejudicam a correspondência exata de k-mers, citar um estudo que meça a queda de desempenho de classificadores (p.ex., o próprio `peresdasilva2024metagenn`, se ele comparar o Kraken2 sob erros de ONT; verificar).] Alignment-based approaches such as BLAST provide a different trade-off between sensitivity and computational cost [9]. [**Nota v2:** [Contribuição] A Introdução motiva o estudo pela sensibilidade do Kraken2 a erros de sequência, mas o Kraken2 nunca é avaliado. Recomenda-se incluí-lo como *baseline* no mesmo conjunto de teste e nos mesmos níveis de ϵ, além de um classificador simples por frequência de k-mers (p.ex., regressão logística ou Naive Bayes sobre perfis de k-mers) como controle de baixo custo. Sem essa comparação, não é possível afirmar que as CNNs são mais robustas que o estado da prática.]

Deep-learning approaches provide an alternative in which discriminative features are learned directly from nucleotide sequences. Neural taxonomic classifiers have been applied to metagenomic and viral classification problems using both flat and hierarchical prediction strategies [10–12]. [**Nota v2:** [Contribuição] Verificar e citar o GeNet (Rojas-Carulla et al., 2019, "GeNet: Deep representations for metagenomics", arXiv), que aplica ResNet a *reads* codificados em *one-hot* com hierarquia taxonômica; é o trabalho mais próximo do presente estudo e um revisor da área provavelmente o mencionará. Explicitar, em seguida, o que este trabalho acrescenta em relação a GeNet, DeepMicrobes, CHEER e MetageNN.] More recently, genomic foundation models such as DNABERT-2 and the Nucleotide Transformer have extended sequence representation learning through large-scale pretraining and token-based genomic representations [13, 14]. In a previous study using the taxonomic dataset underlying the present work, Schoier [15] evaluated pretrained genomic foundation models for fine-grained viral classification and reported substantial difficulty at the species level.

Convolutional neural networks (CNNs) provide a complementary representation strategy. Rather than decomposing DNA into learned or predefined sequence tokens, nucleotide identity can be retained directly at single-base resolution through a multi-channel one-hot tensor. Such representations preserve local sequence order and allow convolutional filters to operate directly over nucleotide positions. CNNs have previously been used to learn predictive sequence features in genomic applications, including regulatory sequence patterns [16–18]. Whether deep 1D convolutional architectures can maintain discriminative performance in a high-cardinality viral taxonomy, and how their learned representations respond to sequence corruption, remains less well characterized.

Here, we evaluate base-level one-hot nucleotide representations using 1D adaptations of ResNet50 and ConvNeXt on a curated closed-set viral classification problem

comprising 13,683 species and 2,702 genera. All target taxa evaluated at inference were represented during training, while individual test fragments were excluded from model optimization. The study therefore addresses discrimination among known taxa rather than open-set recognition or classification of previously unseen Species.

We examine four complementary aspects of model behavior: (i) classification accuracy and computational efficiency on uncorrupted 512-bp fragments; (ii) performance degradation under controlled nucleotide resampling and the effect of noise-aware fine-tuning; (iii) local and global organization of learned latent representations, together with predictive-confidence behavior; and (iv) first-layer filter-derived sequence patterns, matched perturbation effects, exploratory nucleotide-triplet composition, and sliding-window aggregation for variable-length genomic sequences.

2 Methods

[**Nota v2:** Seção de Métodos movida para antes dos Resultados na v2, seguindo o formato usual dos periódicos BMC (Background, Methods, Results, Discussion, Conclusions) e porque os Resultados dependem de definições introduzidas aqui (ϵ e taxa efetiva de troca, nomenclatura dos modelos, protocolo KNN, critérios de validação dos motivos e mecanismo Voter). Confirmar a ordem exigida nas instruções da *BioData Mining*.]

2.1 Taxonomic Curation Pipeline and Hierarchy Reconstruction

The starting dataset used in this study was originally constructed by Schoier [15] for taxonomic classification with DNA foundation models. The dataset was generated from Kraken2 reference libraries [5] through automated sequence acquisition, NCBI taxonomy mapping, removal of ambiguous or unclassified taxa, and class-balanced undersampling. The resulting species- and genus-level datasets contained 13,683 and 2,702 taxonomic classes, respectively.

For the present study, this dataset was further curated to improve taxonomic consistency and reconstruct the hierarchy required for the species–genus–order analyses. Sequence-associated taxonomic identifiers were cross-referenced with the International Committee on Taxonomy of Viruses (ICTV) Master Species List [19] and the NCBI Taxonomy database. [**Nota v2:** [Ref.] Informar a versão (ano) do ICTV Master Species List e a data de acesso ao NCBI Taxonomy, e citar o NCBI Taxonomy (Schoch et al., 2020, *Database*, baa062; não está no `.bib`).] This additional curation was used to resolve synonyms and obsolete nomenclature, recover missing ancestral ranks, and remove ambiguous taxonomic assignments before model training and hierarchical analysis.

We implemented a top-down class-balanced undersampling protocol. The number of sequences per taxonomic node was capped at the size of the smallest eligible class at that rank, yielding a balanced training taxonomy intended to prevent the convolutional models from becoming biased toward overrepresented clades. The resulting dataset comprises a curated taxonomy of 13,683 **Species** and 2,702 **Genera**. [**Nota v2:** Informar: número de fragmentos por classe e total (o conjunto de teste tem 1,706,270 fragmentos, o que sugere ≈17 milhões no total); como os fragmentos de 512 bp foram gerados a partir dos genomas (com ou sem sobreposição, passo utilizado); e quantos genomas de origem há por espécie. Esses dados são necessários para avaliar o risco de fragmentos sobrepostos do mesmo genoma aparecerem em partições diferentes.]

2.2 Dataset Partitioning and Evaluation Setting

The classification task was formulated as a closed-set supervised problem. The same taxonomic label space was represented across the training, validation, and test partitions: all species and genus classes evaluated at test time were also represented during model training. The objective of this setting was to measure discrimination among known viral taxa rather than recognition or assignment of previously unseen species.

The balanced sequence-fragment dataset was partitioned using a stratified 80/10/10 train/validation/test split with a fixed random seed of 42, preserving the class distribution across partitions. Individual sequence fragments were assigned to a single partition, and no exact sequence was shared between the training, validation, and test sets. [**Nota v2:** A ausência de sequências idênticas não exclui fragmentos parcialmente sobrepostos ou quase idênticos (do mesmo genoma ou de genomas muito similares) entre treino e teste, o que pode inflar a acurácia. Esclarecer e, se possível, avaliar também uma partição por genoma de origem.]

The partitioning unit used herewas the sequence fragment, and the reported results should be interpreted as closed-set fragment-level classification within the curated reference distribution. Generalization to previously unseen reference genomes within a known species, as well as open-set recognition of species absent from training, constitutes a distinct evaluation problem and was not assessed in this study.

2.3 Creation of the Long Sequence Dataset

A dedicated **Long Sequence Dataset** was created in order to approximate variable-length long-sequence classification scenarios. While the convolutional networks evaluated here are trained on fixed-length short fragments ($L = 512\,\text{bp}$) to learn local sequence features, long reads produced by third-generation platforms can span several kilobases and exhibit higher per-base error rates [1]. [**Nota v2:** A menção a $L = 1024$ bp foi removida, pois nenhum modelo com essa janela é avaliado no artigo.]

We constructed the Long Sequence Dataset by downloading from NCBI complete, unfragmented genomic sequences (contigs and complete genomes) representing each of the curated viral taxa. [**Nota v2:** Especificar os critérios de inclusão, o número de sequências por espécie, a distribuição de comprimentos e a data de acesso.] These continuous contigs provide the raw substrate for:

- **High-Throughput Windowing:** Applying overlapping windows ($W = 512\,\text{bp}$ with a stride of $S = 256\,\text{bp}$) over the entire contig length to evaluate the sliding-window consensus voting mechanism (`Voter`). This procedure applies the classifiers to long contiguous sequences but does not simulate the error profile of long-read sequencing.
- **Controlled Substitution-Noise Simulation:** Injecting artificial noise dynamically across continuous genomic ranges to study how local spatial features degrade over long, contiguous fragments.

The long-sequence benchmark contained 13,033 of the 13,683 species because complete sequences satisfying the inclusion criteria were unavailable for 650 taxa.

2.4 Out-of-Core Data Engineering and Parquet-Based Processing

Training deep convolutional models (such as ResNet50 and ConvNeXt) on datasets containing tens of millions of genomic sequences can generate memory bottlenecks. We designed and implemented a high-throughput, out-of-core pipeline using the Apache Parquet columnar file format and the Dask distributed computing engine to enable efficient training.

Parallelization was managed with Dask's `LocalCUDACluster`. While training was initiated on an NVIDIA RTX A6000 GPU (48 GB VRAM), the experiments were subsequently run on an NVIDIA RTX PRO 6000 GPU with 96 GB VRAM. [Nota v2: Especificar quais modelos (ou fases) foram treinados em cada GPU; a troca de *hardware* durante o treinamento afeta a reprodutibilidade numérica.] Under this configuration, worker memory usage was capped at 70% to improve execution stability and maintain throughput, allowing for an increase in batch size and parallel loader streams. Two custom PyTorch `IterableDataset` subclasses were created in order to pipe partitioned Parquet files directly into neural network loaders:

1. `DaskCuDFIntegerDataset` **(GPU-Bound):** This dataset class leverages `dask_cudf` to load Parquet partitions directly into GPU VRAM. By executing zero-copy slicing of batched tensors directly in the graphics memory, it minimizes latency overheads associated with transferring data across the PCI-Express bus.
2. `ParquetCPUDataset` **(CPU-Bound):** Implemented using `pandas` and the `pyarrow` engine as a fallback. This CPU-bound loader manages asynchronous data slicing and avoids CUDA context collisions when multiple PyTorch `DataLoader` worker processes access GPU resources simultaneously.

Both dataset classes implement a fast footer-based row-count calculation (`_calculate_total_samples_fast`) whose cost is independent of the number of rows. This method reads only the metadata footer block of the Parquet files to retrieve total row counts without scanning the data.

A **Worker Splitting** routine was designed to guarantee that asynchronous multi-worker processes do not duplicate sequence loads during training. This routine partitions the file paths of the Parquet splits using active worker process IDs, ensuring that each PyTorch worker loads and feeds a completely disjoint, non-overlapping subset of the genomic dataset.

2.5 Spatial Sequence Encoding and Convolutional Architectures

As an alternative to the discrete tokenization characteristic of NLP-based genomic models, this study represents DNA sequences as one-hot nucleotide representations. Raw genomic fragments are sliced into fixed-length windows ($L = 512$ bp) to provide a fixed input length to the convolutional architectures.

2.5.1 One-Hot Nucleotide Encoding

An input DNA sequence $S = (s_1, s_2, \ldots, s_L)$ where $s_i \in \{A, C, G, T\}$ is mapped to a spatial tensor $X \in \mathbb{R}^{4 \times L}$. Each nucleotide is encoded as a four-dimensional one-hot (standard basis) vector:

$$A \mapsto \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad C \mapsto \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad G \mapsto \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad T \mapsto \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \tag{1}$$

`OneHotEmbedding` was implemented as an on-the-fly encoding layer within the model pipeline to minimize disk space and prevent the expansion of stored dataset files. It was integrated as the first step of the neural network graph. A critical detail of our implementation is the treatment of sequence padding (necessary for fragments shorter than $L = 512$ bp or containing IUPAC degenerate bases). The `OneHotEmbedding` layer maps padding indices to a separate vocabulary dimension, performs an expanded PyTorch one-hot operation, and subsequently crops the padding channel. This ensures that regions of sequence padding or empty space are represented strictly by zero-vectors:

$$\text{Padding/Unknown} \mapsto \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{2}$$

2.5.2 The 1D-ResNet50 Architecture

We implemented a 1D model inspired by the ResNet50 architecture [20]. Rather than operating on 2D image coordinates, the Conv2D, BatchNorm2D, and pooling layers were replaced by their 1D counterparts (Conv1D, BatchNorm1D, MaxPool1D). Under this configuration, convolutional filters slide exclusively along the sequence length dimension L, scanning for local spatial features. The network comprises an initial convolutional stem that consists of a Conv1D layer (kernel size 7, stride 2), followed by four residual stages containing $[3, 4, 6, 3]$ bottleneck blocks, respectively. The network's core computational block is the **1D Bottleneck Block** (`BottleneckBlock1D`), which operates with a channel expansion factor of 4. Each block executes a sequence of three 1D convolutions:

1. **Pointwise convolution (kernel size 1):** Compresses the channel dimension to reduce computational complexity.
2. **Spatial convolution (kernel size 3):** Convolves along the spatial sequence axis.
3. **Pointwise convolution (kernel size 1):** Projects the feature map back to the expanded channel space, establishing the output dimension.

These convolutions are intercalated with 1D Batch Normalization [21] and ReLU activation functions. The output of the third convolution is summed with the block's input via a residual identity shortcut connection, facilitating gradient propagation through

the depth of the network. [Nota v2: Informar se o *stem* do ResNet50 inclui BatchNorm, ReLU e MaxPool (como descrito para o ConvNeXt), as larguras de canal de cada estágio e a dimensão do vetor latente antes da camada final.] Following the final residual stage, global max pooling aggregates each feature channel across the sequence dimension. The resulting feature vector is flattened, regularized using Dropout (p = 0.3), and projected to the target taxonomic classes through a fully connected layer.

2.5.3 The 1D-ConvNeXt Architecture

A 1D ConvNeXt-inspired architecture was developed based on the ConvNeXt design proposed by Liu et al. [22]. Our implementation adapts the ConvNeXt block to one-dimensional genomic sequences while preserving the ResNet50 stage configuration ([3,4,6,3]) and a ResNet-style initial downsampling stem to facilitate architectural comparison.

Initial Stage and Preserved Pooling:

The model accepts the $4 \times L$ tensor and passes it through an initial Conv1D layer (kernel size 7, stride 2) followed by BatchNorm1D, ReLU, and MaxPool1D (stride 2). This layout corresponds to a ResNet-style downsampling stem. Although atypical for standard ConvNeXt models, which use a non-overlapping "patchify" stem, this downsampling stem is preserved in our graph to maintain architectural parity and identical feature dimensions for head-to-head comparison with the ResNet50 baseline.

Four-Stage Residual Backbone:

The main body of the network consists of four hierarchical convolutional stages containing $[3, 4, 6, 3]$ blocks, respectively. Within each stage, spatial and channel-wise features are processed through 1D ConvNeXt Blocks. Each block comprises two parallel paths:

- **Shortcut Path A:** An identity connection that copies the input tensor (leveraging a 1D LayerNorm downsampling block when transitioning between stages to adjust channel dimensions and resolution).
- **Convolutional Path B:** Performs spatial and channel mixing using depthwise and pointwise operations:
  1. **1D Depthwise Convolution:** Uses a large spatial kernel (kernel size 7 with stride 1 within stages and stride 2 in stage-transition blocks, and `groups=channels`) to mix spatial information along the sequence length axis independently for each feature channel. The large kernel increases the effective receptive field while maintaining computational efficiency.
  2. **LayerNorm1D:** Applies normalization across the channel dimension (replacing Batch Normalization with Layer Normalization, consistent with the ConvNeXt design).
  3. **Pointwise Expansion (1×1 Linear):** Expands the channel dimension by a factor of 4.
  4. **GELU Activation:** Replaces traditional ReLU with the Gaussian Error Linear Unit for smooth non-linear gating.

5. **Pointwise Projection (1×1 Linear):** Projects the expanded representation to the output channel dimension required by the residual stage.
6. **Layer Scale:** Applies a diagonal scaling matrix initialized with $\gamma = 1 \times 10^{-6}$ to stabilize the training of deep networks.

The outputs of Path A and Path B are summed to produce the final block output.

Taxonomic Classification Head:

We replaced the conventional Global Average Pooling (GAP) layer with Global Max Pooling (GMP), allowing the classifier to retain the strongest activation from each feature channel rather than averaging responses across the entire sequence. The pooled feature tensor is passed through a Dropout layer ($p = 0.3$) and a Fully Connected (Linear) layer to project the deep features into the high-cardinality target space representing the 13,683 **species** or 2,702 **genera** taxonomic classes.

2.6 Nucleotide Corruption and Consensus Inference Mechanism

[**Nota v2:** [Coerência] Subseção movida na v2 para antes da estratégia de treinamento, porque o ajuste com ruído (*noise-aware fine-tuning*) utiliza o procedimento de corrupção aqui definido.]

Controlled nucleotide corruption was applied to the input sequences to evaluate classifier robustness to local sequence perturbations. Although motivated by sequencing-error profiles reported for Oxford Nanopore sequencing, the procedure models independent nucleotide resampling only and should therefore be interpreted as a controlled robustness stress test rather than as a complete simulation of ONT sequencing errors. More realistic long-read simulators model empirical error distributions, including insertions and deletions [7, 8].

For each sequence window $X \in \{0, 1, 2, 3\}^L$ of length $L = 512$ bp, a Bernoulli mask was independently sampled at each position:

$$M_i \sim \text{Bernoulli}(\epsilon), \tag{3}$$

where $\epsilon \in \{0.00, 0.02, 0.05, 0.10, 0.15\}$ denotes the probability that a nucleotide position is selected for resampling. For positions satisfying $M_i = 1$, a nucleotide was independently sampled from the four-symbol alphabet,

$$R_i \sim \text{Uniform}(\{0, 1, 2, 3\}), \tag{4}$$

and the perturbed sequence was defined as

$$\tilde{X}_i = \begin{cases} R_i, & M_i = 1, \\ X_i, & M_i = 0. \end{cases} \tag{5}$$

Because the resampled nucleotide can coincide with the original nucleotide, ϵ represents the positional resampling probability rather than the effective substitution rate. Under uniform four-base resampling, a selected position changes nucleotide with probability 3/4. Consequently, the expected fraction of positions that effectively differ from the original sequence is

$$P(\tilde{X}_i \neq X_i) = \frac{3}{4}\epsilon. \tag{6}$$

The nominal resampling levels of 2%, 5%, 10%, and 15% therefore correspond to expected effective nucleotide-change rates of approximately 1.5%, 3.75%, 7.5%, and 11.25%, respectively. Corruption was generated stochastically during evaluation, producing independent perturbed realizations at each evaluated resampling level. Because this procedure does not introduce insertions or deletions, sequence length remained unchanged.

2.6.1 Sliding-Window Consensus and Confidence Aggregation

We developed the **Voter** sliding-window consensus mechanism to apply our spatial models, trained on $L = 512\,\text{bp}$ fragments, to contiguous variable-length sequences without assembly.

Given a contig $\mathbf{G}$ of length M, overlapping windows of size $W = 512\,\text{bp}$ are extracted with stride $S = 256\,\text{bp}$, yielding

$$N = \left\lfloor \frac{M - W}{S} \right\rfloor + 1 \tag{7}$$

windows $\mathbf{w}_j = \mathbf{G}[jS : jS + W]$, for $j = 0, \ldots, N-1$. The final $M - (N-1)S - W$ nucleotides, which do not complete a window, are not evaluated. [Nota v2: Informar como foram tratadas sequências com $M < W$ e o critério de desempate da votação majoritária.] Each window is transformed by the custom `OneHotEmbedding` layer, with IUPAC degenerate bases and padding mapped to null vectors, and processed by the 1D-CNN f_θ to obtain logits $\mathbf{z}_j \in \mathbb{R}^C$. These are converted to class probabilities using Softmax:

$$p_{j,c} = \frac{\exp(z_{j,c})}{\sum_{k=1}^{C} \exp(z_{j,k})}. \tag{8}$$

We compared two aggregation strategies: hard majority voting and mean-logit aggregation. Hard voting assigns each window to its most probable class and selects the modal prediction across all windows:

$$\hat{y}_{\text{hard}} = \text{mode}\left\{ \arg\max_c p_{j,c} \right\}_{j=0}^{N-1}. \tag{9}$$

For mean-logit aggregation, the pre-Softmax output vectors produced for all windows were averaged:

$$\bar{\mathbf{z}} = \frac{1}{N} \sum_{j=0}^{N-1} \mathbf{z}_j, \tag{10}$$

and the final prediction was obtained from the class with the highest aggregated logit:

$$\hat{y}_{\text{mean-logit}} = \arg\max_c \bar{z}_c. \tag{11}$$

This strategy combines class evidence across overlapping sequence windows before the Softmax transformation, while majority voting considers only the highest-ranked class of each individual window.

2.7 Training Strategy and Optimization Phases

A customized multi-phase optimization schedule was designed for both the **ResNet50** and **ConvNeXt** architectures to address the optimization challenges of high-cardinality classification. Hyperparameters were selected empirically based on validation performance. [**Nota v2:** Informar o tamanho de lote do treinamento principal, o critério de seleção do *checkpoint* final (p.ex., menor perda de validação) e se o *backbone* permaneceu congelado durante o *warmup* dos modelos treinados do zero. Notar que os orçamentos de treinamento diferem entre modelos (p.ex., 154 épocas para o ResNet50 de espécie, 95 para o ConvNeXt do zero e 117 para o ConvNeXt por transferência, além das 133 épocas do modelo de gênero), o que é um fator de confusão na comparação entre arquiteturas e deve constar nas Limitações.]

[**Nota v2:** [Ref.] Vários métodos usados não têm referência: AdamW (Loshchilov e Hutter, ICLR 2019), *label smoothing* (Szegedy et al., CVPR 2016), *cosine annealing* com *warm restarts* (Loshchilov e Hutter, ICLR 2017), *one-cycle* (Smith e Topin, 2019), Layer Normalization (Ba et al., 2016), GELU (Hendrycks e Gimpel, 2016), *layer scale* (Touvron et al., ICCV 2021), precisão mista (Micikevicius et al., ICLR 2018), *dropout* (Srivastava et al., JMLR 2014), PyTorch (Paszke et al., NeurIPS 2019) e Dask (Rocklin, SciPy 2015). Nenhuma dessas entradas está no `.bib`.] All training runs used the AdamW optimizer, gradient-norm clipping with a threshold of 1.0, and automatic mixed precision (AMP). The default objective function across all experiments was the standard categorical Cross-Entropy Loss:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{e^{\mathbf{z}_{i,y_i}}}{\sum_{j=1}^{C}e^{\mathbf{z}_{i,j}}} \tag{12}$$

where $\mathbf{z}$ represents the output logit tensor, y is the ground-truth taxonomic class, N is the batch size, and C represents the total number of target classes. In specific high-complexity species training phases, we modified this objective to include Label Smoothing ($\alpha = 0.1$) to penalize overconfidence in the 13,683-class classification head:

$$\mathcal{L}_{\text{CE, smooth}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{C} q_{i,k}\log\frac{e^{\mathbf{z}_{i,k}}}{\sum_{j=1}^{C}e^{\mathbf{z}_{i,j}}}, \quad q_{i,k} = \begin{cases} 1-\alpha+\frac{\alpha}{C} & \text{if } k = y_i \\ \frac{\alpha}{C} & \text{otherwise} \end{cases} \tag{13}$$

The multi-phase schedules used in this project were built around three core stages:

1. **Warmup Phases:** This initial stage uses a low learning rate to stabilize the early optimization of the newly initialized classification head [23]. [**Nota v2:** [Ref.] Removida a afirmação de que a instabilidade inicial é "particularmente sensível em problemas de granularidade taxonômica extrema": Gotmare et al. analisam o *warmup* em geral, e `toledo2020case` (*overfitting* em classificadores multiclasse) não trata de *warmup* nem de alta cardinalidade. Se quiser manter a afirmação, é necessária uma referência específica. Atualizar também `gotmare2018closer` para a versão publicada (ICLR 2019).] [**Nota v2:** [Ref.] `toledo2020case` trata de *overfitting* em classificadores multiclasse e não sustenta especificamente o uso de *warmup*; verificar a pertinência. `gotmare2018closer` está citado como *preprint* do arXiv, mas foi publicado no ICLR 2019; atualizar a entrada. A chave `toledo2020case` indica 2020, mas a entrada tem `year = 2019`.]

2. **Optimization Phase:** In this primary learning stage, the entire convolutional backbone is trainable, allowing all layers to update their representations. The learning rate is raised to its peak value (typically 3×10^{-5} to 1×10^{-4}) and smoothly decayed following a `CosineAnnealingLR` schedule.
3. **Exploratory Fine-Tuning Phases:** These phases were intended to explore different regions of the loss landscape. `CosineAnnealingWarmRestarts` paired with Label Smoothing ($\alpha = 0.1$) were used to periodically increase the learning rate and allow additional exploration during fine-tuning.

The individual model training runs were organized into structured, sequential schedules detailed below.

2.7.1 ResNet50 Genus-Level Training Schedule

The 1D adaptation of ResNet50 was trained at the genus level (C = 2,702 classes) across a sequential four-phase schedule that follows the three-stage framework described above:

- **Phase 1 (Warmup):** Trained for 13 epochs using a `OneCycleLR` scheduler with a peak learning rate of $\eta_{\max} = 5 \times 10^{-6}$, weight decay of 0.07, and pct_start = 0.3.
- **Phase 2 (Initial Optimization):** Unfrozen backbone trained for 40 epochs under a `CosineAnnealingLR` scheduler ($\eta_{\max} = 3 \times 10^{-5}$, weight decay 0.07, and terminal $\eta_{\min} = 1 \times 10^{-7}$).
- **Phase 3 (Main Optimization):** Unfrozen backbone trained for 40 epochs under `CosineAnnealingLR` with a higher boundary ($\eta_{\max} = 1 \times 10^{-4}$, $\eta_{\min} = 1 \times 10^{-7}$, weight decay 0.07).
- **Phase 4 (Fine-Tuning with Warm Restarts):** Refined for 64 epochs under cyclic `CosineAnnealingWarmRestarts` ($\eta_{\max} = 5 \times 10^{-5}$, $\eta_{\min} = 1 \times 10^{-7}$, weight decay 0.07).

2.7.2 ResNet50 Species-Level Training Schedule (Ab Initio)

Because repeated attempts at transfer learning from the ResNet50 genus backbone to the species classification head did not converge stably, the species-level ResNet50 model was trained from scratch (C = 13,683 classes) using a parallel four-stage schedule:

- **Phase 1 (Warmup):** Trained for 10 epochs under a `OneCycleLR` scheduler ($\eta_{\max} = 5 \times 10^{-6}$, weight decay 0.07).
- **Phase 2 (Initial Optimization):** Trained for 40 epochs under `CosineAnnealingLR` ($\eta_{\max} = 3 \times 10^{-5}$, weight decay 0.07, $\eta_{\min} = 1 \times 10^{-7}$).
- **Phase 3 (Main Optimization):** Trained for 40 epochs under `CosineAnnealingLR` ($\eta_{\max} = 1 \times 10^{-4}$, $\eta_{\min} = 1 \times 10^{-7}$, weight decay 0.07).
- **Phase 4 (Fine-Tuning with Warm Restarts):** Trained for 64 epochs using a cyclic `CosineAnnealingWarmRestarts` scheduler ($\eta_{\max} = 5 \times 10^{-5}$, $\eta_{\min} = 1 \times 10^{-7}$, weight decay 0.07).

2.7.3 ConvNeXt Genus-Level Training Schedule

The 1D ConvNeXt architecture was trained on the 2,702 genus classes using the following non-cyclic schedule:

- **Phase 1 (Warmup):** Trained for 13 epochs using `OneCycleLR` ($\eta_{\max} = 5 \times 10^{-6}$, weight decay 0.07, pct_start = 0.3).
- **Phase 2 (Initial Optimization):** Trained for 40 epochs under `CosineAnnealingLR` ($\eta_{\max} = 3 \times 10^{-5}$, weight decay 0.07, $\eta_{\min} = 1 \times 10^{-7}$).
- **Phase 3 (Main Optimization):** Trained for 40 epochs under `CosineAnnealingLR` ($\eta_{\max} = 1 \times 10^{-4}$, weight decay 0.07, $\eta_{\min} = 1 \times 10^{-7}$).
- **Phase 4 (Final Fine-Tuning):** Trained for 40 epochs using a conservative `CosineAnnealingLR` decay curve ($\eta_{\max} = 1 \times 10^{-6}$, weight decay 0.07, $\eta_{\min} = 1 \times 10^{-7}$) for final low-learning-rate optimization.

2.7.4 ConvNeXt Species-Level Training Schedule (Ab Initio)

A customized **five-phase, double-warmup schedule** incorporating label smoothing was used:

- **Phase 1 (Initial Warm-up):** Trained for 10 epochs using `OneCycleLR` ($\eta_{\max} = 1 \times 10^{-4}$, weight decay 0.07).
- **Phase 2 (Secondary Warm-up):** Trained for 15 epochs using `OneCycleLR` at a lower peak ($\eta_{\max} = 5 \times 10^{-6}$, weight decay 0.07) to provide a gradual transition before the main optimization phase.
- **Phase 3 (Main Optimization):** Trained for 30 epochs using `OneCycleLR` with an aggressive peak learning rate ($\eta_{\max} = 4 \times 10^{-3}$, weight decay 0.05) under a smoothed loss function (`Label Smoothing = 0.1`). This higher learning rate was intended to encourage broader exploration of the optimization landscape during the early stages of training.
- **Phase 4 (Learning Rate Annealing):** Trained for 10 epochs using `CosineAnnealingLR` ($\eta_{\max} = 1 \times 10^{-4}$, $\eta_{\min} = 1 \times 10^{-7}$, weight decay 0.07) with `Label Smoothing = 0.1`.
- **Phase 5 (Fine-Tuning with Warm Restarts):** Trained for 30 epochs using `CosineAnnealingWarmRestarts` ($\eta_{\max} = 1 \times 10^{-3}$, $\eta_{\min} = 1 \times 10^{-7}$, weight decay 0.07, and a cycle length of 10 epochs) under `Label Smoothing = 0.1`.

2.7.5 ConvNeXt Species-Level Transfer Learning

We developed a **six-phase transfer learning schedule** starting from the genus-level ConvNeXt weights:

- **Phase 1 (Classifier Adaptation):** The pre-trained genus-level backbone was completely frozen. Only the 13,683-class species linear layer was optimized for 8 epochs without a scheduler ($\eta_{\max} = 1 \times 10^{-3}$, weight decay 0.07).
- **Phase 2 (Backbone Fine-Tuning):** All layers were completely unfrozen. The network was trained for 13 epochs using `CosineAnnealingLR` ($\eta_{\max} = 1 \times 10^{-5}$, $\eta_{\min} = 1 \times 10^{-7}$) with a high weight decay coefficient (0.15) to regularize the backbone after unfreezing.

- **Phase 3 (Learning Rate Annealing):** Trained for 7 epochs using `CosineAnnealingLR` ($\eta_{\max} = 3 \times 10^{-5}$, $\eta_{\min} = 1 \times 10^{-7}$, weight decay 0.07).
- **Phase 4 (Warm-Restart Fine-Tuning):** Trained for 40 epochs using `CosineAnnealingWarmRestarts` ($\eta_{\max} = 1 \times 10^{-4}$, $\eta_{\min} = 1 \times 10^{-7}$, weight decay 0.07).
- **Phase 5 (Warm-Restart Fine-Tuning with Label Smoothing):** Trained for 30 epochs under `CosineAnnealingWarmRestarts` with `Label Smoothing = 0.1` ($\eta_{\max} = 1 \times 10^{-3}$, $\eta_{\min} = 1 \times 10^{-7}$, weight decay 0.07, and cycle length of 10 epochs) intended to improve discrimination among closely related species.
- **Phase 6 (Final Convergence):** Concluded with 19 final epochs of `CosineAnnealingWarmRestarts` with `Label Smoothing = 0.1` ($\eta_{\max} = 1 \times 10^{-4}$, $\eta_{\min} = 1 \times 10^{-7}$, weight decay 0.07).

2.7.6 Noise-Aware Fine-Tuning

Noise-aware fine-tuning was applied to the species-level ResNet50 and ConvNeXt models. ResNet50 was initialized from its best clean-trained species checkpoint and ConvNeXt was initialized from the species classifier obtained through genus-to-species transfer learning. All network parameters remained trainable during this stage.

Controlled nucleotide resampling was generated online during both training and validation using the same corruption procedure described in Section 2.6. Each nucleotide position was selected for resampling independently with probability $\epsilon = 0.05$. Because the replacement nucleotide was sampled uniformly from all four canonical bases, including the original nucleotide, this corresponds to an expected effective nucleotide-change rate of $0.75 \times 0.05 = 0.0375$ (3.75%). During fine-tuning, ϵ was fixed rather than sampled from a range.

Both architectures were fine-tuned for 15 epochs using AdamW with an initial learning rate of 5×10^{-5} and weight decay of 1×10^{-4}. The objective was categorical cross-entropy with label smoothing $\alpha = 0.1$. A CosineAnnealingLR scheduler was used with a minimum learning rate of 1×10^{-7}. Training used batches of 4,096 sequences and automatic mixed-precision optimization.

A checkpoint was stored after each epoch together with the model, optimizer, learning-rate scheduler, and gradient-scaler states. The most recent checkpoint was additionally maintained to support training resumption.

2.8 Latent Space Extraction and Cluster Analysis

To investigate the internal representations learned by the neural classifier, latent feature vectors were extracted from the trained 1D-CNN prior to the final classification layer. For each input sequence, the activation of the final feature representation was retained as a fixed-dimensional embedding, thereby representing the sequence in the model's learned feature space rather than in the original nucleotide space. These latent representations were subsequently used to assess whether the classifier learned separable sequence representations corresponding to the underlying taxonomic structure.

Because the resulting feature vectors are high-dimensional, Uniform Manifold Approximation and Projection (UMAP) was used to obtain a two-dimensional representation for visualization and exploratory analysis. [**Nota v2:** [Ref.] O UMAP não está citado e não há entrada no `.bib`; incluir McInnes, Healy e Melville (2018), arXiv:1802.03426.] UMAP constructs a neighborhood graph in the original feature space and projects the resulting structure into a lower-dimensional manifold while attempting to preserve local relationships between samples. The same dimensionality-reduction procedure was applied consistently across the evaluated models to enable qualitative comparison of the learned representations.

In addition to visual inspection of the two-dimensional embeddings, quantitative cluster-separation metrics were calculated to characterize the organization of the latent space. The Silhouette Score [24] was used to quantify global class separation by comparing within-class cohesion with separation from neighboring classes. Where applicable, complementary cluster-quality measures were used to evaluate the compactness and separation of taxonomic groups. [**Nota v2:** Nenhuma outra métrica de *cluster* é reportada nos Resultados; especificar quais foram usadas ou remover esta frase.] These analyses were performed on the extracted latent representations rather than on the model's final class probabilities, allowing the organization of the learned feature space to be assessed independently of the final decision layer.

Because no single metric fully characterizes the quality of a genomic embedding space, we evaluated complementary global and local properties of the learned representations, consistent with recent methodological work on the evaluation of genomic vector representations [25]. [**Nota v2:** [Ref.] `zheng2024evaluating` trata de *embeddings* não supervisionados de *regiões genômicas* (intervalos), não de *embeddings* de sequências aprendidos por classificadores supervisionados. A redação "consistent with" é adequada como analogia; evitar apresentá-lo como validação direta do protocolo.]

2.8.1 Quantitative Latent-Space Analysis

All quantitative latent-space analyses were performed directly on the original high-dimensional feature representations extracted immediately before the final linear classification layer. These pre-classifier embeddings were used without dimensionality reduction for all quantitative evaluations. UMAP projections were used exclusively for qualitative visualization and were not involved in the computation of KNN Accuracy or Silhouette Scores.

Local taxonomic consistency was evaluated using a k-nearest-neighbor classifier with $k = 5$. For each model, the complete set of embeddings extracted from the held-out test set ($N = 1{,}706{,}270$) was randomly partitioned using a stratified 80/20 reference-query split into an 80% reference set and a 20% query set using a fixed random seed of 42. The two partitions were disjoint, preventing a query embedding from being included among its own nearest neighbors.

Before neighbor search, reference and query embeddings were L_2-normalized. Similarity between embeddings was evaluated using cosine distance,

$$d_{\cos}(\mathbf{x}, \mathbf{y}) = 1 - \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2}. \tag{14}$$

[**Nota v2:** Referência e consulta vêm do mesmo conjunto de teste; fragmentos do mesmo genoma de origem (possivelmente sobrepostos) podem ser vizinhos entre si, inflando a consistência local. Sugiro repetir a análise com divisão

referência/consulta disjunta por genoma de origem.] For each query embedding, the five nearest embeddings in the reference set were retrieved, and its neighborhood-derived taxonomic label was assigned by majority vote among the corresponding reference labels. KNN Accuracy was defined as the proportion of query embeddings for which this predicted label matched the ground-truth taxonomic label. The same procedure was applied independently at the species, genus, and order levels. Nearest-neighbor search and cosine-distance computation were implemented in PyTorch and executed on the GPU.

Global class separation was evaluated using the Silhouette Score in the same original high-dimensional embedding space with cosine distance. Because exact Silhouette computation over the complete test set is computationally prohibitive, up to 50,000 embeddings were randomly sampled for each model-level evaluation. Taxonomic classes represented by fewer than two samples in the resulting subset were excluded from the Silhouette calculation. [**Nota v2:** Com até 50,000 amostras para 13,683 espécies (≈3,7 por classe), muitas espécies terão menos de duas amostras ou estimativas muito ruidosas. Informar quantas classes foram efetivamente incluídas em cada nível e considerar amostragem estratificada por classe.] In addition to the global score, per-taxon Silhouette Scores were obtained by averaging the sample-level Silhouette values associated with each taxonomic class.

2.8.2 Parent–Child Taxonomic Correspondence

Parent–child correspondence was examined for species→genus and genus→order relationships. The curated taxonomic hierarchy was used to map each child taxon to its corresponding parent taxon.

For each parent taxon, the metric values of its associated child taxa were summarized using an unweighted arithmetic mean. Thus, for a parent taxon p with a set of child taxa $\mathcal{C}_p$, the aggregated child-level metric was defined as

$$\overline{M}_{\text{child},p} = \frac{1}{|\mathcal{C}_p|} \sum_{c \in \mathcal{C}_p} M_c, \tag{15}$$

where M_c denotes either the KNN Accuracy or Silhouette Score of child taxon c. This aggregated child-level value was then compared with the independently calculated value of the same metric for the corresponding parent taxon.

Hierarchical correspondence was quantified using both Pearson's correlation coefficient (r), to measure linear association, and Spearman's rank correlation coefficient (ρ), to measure monotonic correspondence between child-aggregated and parent-level metric values.

KNN Accuracy and Silhouette Score were analyzed independently. For the KNN analysis, only parent taxa for which both child-level KNN Accuracy and parent-level KNN Accuracy were available were retained. The Silhouette analysis used its own set of valid parent taxa, requiring both child-level and parent-level Silhouette Scores. Consequently, missing Silhouette estimates did not exclude otherwise valid taxa from the KNN correspondence analysis, and the number of parent taxa included in each comparison could differ between the two metrics and across model architectures.

Because metric availability differed between KNN Accuracy and Silhouette Score, the number of parent taxa included in each correspondence analysis was recorded separately. For the species→genus comparison, KNN-based correspondence included 2,321 genera for all model architectures, and for the Silhouette-based analysis it included between 2,154 and 2,166 genera, depending on the model. For the genus→order comparison, KNN-based correspondence included all 74 orders for every model, while the Silhouette-based analysis included 73 orders for all models except the ConvNeXt model trained from scratch, for which all 74 orders were available. These metric-specific sets were analyzed independently rather than restricting both analyses to their intersection.

2.8.3 Visualization with UMAP

For qualitative visualization of the learned latent representations, the high-dimensional feature vectors extracted from the held-out test set were projected into two dimensions using Uniform Manifold Approximation and Projection (UMAP). The projection was computed using $n_{\text{neighbors}} = 15$, min_dist $= 0.1$, two output dimensions, and cosine distance, with a fixed random seed of 42.

UMAP was applied to the complete set of test embeddings (N = 1,706,270 per model), without additional subsampling. No explicit L_2 normalization was applied to the feature vectors before UMAP projection.

[**Nota v2:** Removido na v2 um parágrafo que repetia, pela terceira vez nesta seção, que o UMAP foi usado apenas para visualização.]

2.9 Explainability and Biological Feature Extraction

2.9.1 CNN Motif Extraction and Matched Perturbation Analysis

Sequence motifs encoded by the CNNs were extracted from the first `Conv1D` layer of each trained model. First-layer convolutional filters were treated as candidate position-specific sequence detectors, following the established use of CNN filters for genomic motif representation [16, 18]. For each of the seven filter positions, the nucleotide associated with the largest signed filter weight was selected, without taking the absolute value of the weights (`use_abs=False`), yielding one consensus 7-mer per filter. Duplicate consensus sequences were collapsed before subsequent analysis.

Occurrences of each consensus motif were identified by exact 7-mer matching in the held-out test sequences. Only sequences containing at least one exact occurrence of the motif and admitting a valid matched control region were included. Within each Monte Carlo repetition, one motif occurrence was randomly selected from each eligible sequence. A matched control region of seven nucleotides was independently sampled from the same sequence. Control regions were required to contain only valid A, C, G, and T nucleotides and could not overlap any occurrence of the corresponding motif in that sequence.

The selected motif and matched random region were perturbed in parallel. Each nucleotide in the selected region was replaced by a different nucleotide using a randomly generated offset from $\{1, 2, 3\}$ under the integer A/C/G/T encoding. The same set of substitution offsets was applied to the motif and its matched random region, ensuring that the two perturbations differed only in their genomic location rather than

in the substitution pattern itself. The procedure was repeated for five Monte Carlo repetitions using a fixed random seed of 42.

For sequence i and repetition r, the effect of motif perturbation was defined as the decrease in probability assigned to the ground-truth class,

$$\Delta p^{(r)}_{\text{motif},i} = p^{\text{baseline}}_{\text{true},i} - p^{\text{motif},(r)}_{\text{true},i}, \tag{16}$$

where the corresponding matched random-region effect was

$$\Delta p^{(r)}_{\text{random},i} = p^{\text{baseline}}_{\text{true},i} - p^{\text{random},(r)}_{\text{true},i}. \tag{17}$$

The effects were averaged across the five Monte Carlo repetitions for each sequence–motif pair,

$$\Delta p_{\text{motif},i} = \frac{1}{R}\sum_{r=1}^{R} \Delta p^{(r)}_{\text{motif},i}, \qquad \Delta p_{\text{random},i} = \frac{1}{R}\sum_{r=1}^{R} \Delta p^{(r)}_{\text{random},i}, \tag{18}$$

with $R = 5$. The motif-specific perturbation effect was then defined as

$$\Delta p_{\text{specific},i} = \Delta p_{\text{motif},i} - \Delta p_{\text{random},i}. \tag{19}$$

Positive values of $\Delta p_{\text{specific}}$ indicate that perturbing the learned motif reduced the true-class probability more strongly than perturbing a matched random region of the same sequence.

Statistical evidence for motif-specific predictive contribution was evaluated independently for each model using a paired, one-sided Wilcoxon signed-rank test [Nota v2: [Ref.] Citar Wilcoxon (1945) e, para o controle de FDR, Benjamini e Hochberg (1995); nenhuma das duas está no `.bib`.] comparing Δp_{motif} and Δp_{random} across sequences, with the alternative hypothesis $\Delta p_{\text{motif}} > \Delta p_{\text{random}}$. The resulting p-values were corrected for multiple motif testing using the Benjamini–Hochberg false discovery rate (BH-FDR) procedure at $\alpha = 0.05$. A motif was considered statistically validated when its FDR-adjusted p-value satisfied $p_{\text{FDR}} < 0.05$, its mean motif-specific effect was positive ($\overline{\Delta p}_{\text{specific}} > 0$), and it was supported by at least five distinct test sequences ($N \geq 5$).

2.9.2 Triplet-Composition Association of Learned Motifs

Downstream exploratory sequence-composition analyses were performed using the statistically validated motifs identified by the matched perturbation analysis. Since the genomic fragments we evaluated were not restricted to annotated coding sequences and lacked biological reading frames, this analysis should not be interpreted as a direct measure of translational codon usage.

We reconstructed nucleotide sequences from 50 test set batches [Nota v2: Informar o tamanho dos lotes, isto é, o número de fragmentos usados como fundo.] and scanned them in a fixed frame (frame 0) to extract non-overlapping triplets. We grouped these triplet counts using the standard genetic code and normalized them with the Relative Synonymous Codon Usage (RSCU) formula. [Nota v2: [Ref.] Citar a definição original do RSCU (Sharp, Tuohy e Mosurski, 1986, *Nucleic Acids Res.* 14:5125–5143); `plotkin2011synonymous`, já no `.bib` e não citado, pode contextualizar o viés de códons.] In this context, we use RSCU simply to normalize triplet composition, rather

than to infer biological codon preferences. The background metric was calculated as follows:

$$RSCU_i = \frac{n_i}{\frac{1}{m_a}\sum_{j=1}^{m_a} n_j}, \tag{20}$$

where n_i is the observed count of nucleotide triplet i, m_a is the number of synonymous codons for its encoded amino acid, and the summation is over all synonymous codons for that amino acid. Codons belonging to amino acids with no observations were assigned an RSCU value of 1.0, representing neutral expected usage.

For each model, the validated motif set was decomposed into overlapping nucleotide triplets. For a motif of length K, all consecutive triplets $\mathbf{m}[i:i{+}3]$, with $i = 0, \ldots, K-3$, were extracted and matched against the 64 canonical nucleotide triplets. Motifs containing non-nucleotide or padding characters were excluded. The frequency of each triplet within the motif set was calculated as

$$F_i = \frac{n_i^{\text{motif}}}{\sum_{k=1}^{64} n_k^{\text{motif}}}, \tag{21}$$

and compared with its corresponding global RSCU value.

Pearson and Spearman correlation coefficients were calculated between the global RSCU values and motif-level triplet frequencies to quantify the degree of linear and rank-based association between the learned sequence patterns and the background triplet-composition profile.

2.9.3 Motif Co-occurrence and Triplet-Composition Similarity

A co-occurrence matrix was constructed from the identified motif occurrences. Each sequence was represented by a binary vector indicating the presence or absence of each statistically validated motif. For every pair of motifs i and j, co-occurrence was quantified using the Jaccard index:

$$J_{ij} = \frac{|M_i \cap M_j|}{|M_i \cup M_j|}, \tag{22}$$

where M_i and M_j denote the sets of sequences containing motifs i and j, respectively. Motif pairs were ranked by their Jaccard scores, and the top 200 interactions were selected for downstream analysis.

We calculated the mean RSCU value of the codons represented within each motif and compared these values between paired motifs in order to check whether co-occurring motifs also shared RSCU-derived triplet-composition characteristics. For each selected interaction (i, j), the corresponding motif-level triplet-composition profiles were summarized as $\mathrm{R\bar{S}CU}_i$ and $\mathrm{R\bar{S}CU}_j$. The association between the triplet composition of paired motifs was evaluated using Pearson correlation:

$$r = \mathrm{corr}\left(\mathrm{R\bar{S}CU}_i, \mathrm{R\bar{S}CU}_j\right). \tag{23}$$

The resulting co-occurrence and codon-usage relationships were visualized to identify motif pairs showing both strong sequence-level association and concordant RSCU-derived composition. This analysis was used to determine whether the

sequence features learned by the CNN occur independently or form coordinated motif combinations associated with shared triplet-composition patterns.

2.9.4 Hierarchical Entropy Analysis

A hierarchical entropy analysis was performed on the species-level output distributions of each model to characterize predictive uncertainty across taxonomic levels. For each input sequence, species logits were converted to probabilities using the Softmax function. Species-level uncertainty was then quantified using Shannon entropy:

$$H_{\text{species}} = -\sum_{c=1}^{C} p_c \log p_c, \tag{24}$$

where p_c denotes the predicted probability of species class c and C is the total number of species classes.

To determine whether predictive uncertainty was concentrated among taxonomically related species, species probabilities belonging to the same genus were summed to obtain a genus-level probability distribution:

$$p_g = \sum_{c \in \mathcal{S}_g} p_c, \tag{25}$$

where $\mathcal{S}_g$ denotes the set of species classes assigned to genus g. Genus-level entropy was subsequently calculated as

$$H_{\text{genus}} = -\sum_{g=1}^{G} p_g \log p_g, \tag{26}$$

where G is the number of genus classes.

The hierarchical entropy gap was defined as

$$\Delta H = H_{\text{species}} - H_{\text{genus}}. \tag{27}$$

To characterize the distribution of internal feature activations, the pre-classifier latent representation was captured using a forward pre-hook immediately before the final linear classification layer. For each sequence, the latent vector $\mathbf{z} \in \mathbb{R}^D$ was flattened and converted into a non-negative activation-magnitude distribution:

$$q_j = \frac{|z_j|}{\sum_{k=1}^{D} |z_k|}, \tag{28}$$

where D denotes the dimensionality of the pre-classifier representation.

Activation entropy was then calculated as

$$H_{\text{activation}} = -\frac{1}{\log D} \sum_{j=1}^{D} q_j \log q_j. \tag{29}$$

Normalization by $\log D$ bounds $H_{\text{activation}}$ between 0 and 1, with lower values indicating that activation magnitude is concentrated in a smaller subset of latent dimensions and higher values indicating a more distributed activation profile.

Species entropy, genus entropy, normalized activation entropy, and the hierarchical entropy gap were computed for all test batches and averaged across the held-out test set for each model.

3 Results

Throughout this section and the Discussion, the labels ResNet50 and ConvNeXt denote complete trained configurations, comprising architecture, training schedule, regularization (including label smoothing), and initialization, rather than the architectures in isolation. Because these factors were not varied independently, differences between configurations are reported descriptively and are not attributed to architecture alone.

3.1 Taxonomic Classification Performance on Uncorrupted Short Sequences ($L = 512$ bp)

On uncorrupted 512-bp fragments, ResNet50 achieved the best classification performance, although the magnitude of its advantage over ConvNeXt depended strongly on taxonomic granularity (Table 1). At the genus level, the two architectures performed similarly, with only a small difference in Weighted F1-score (86.19% for ResNet50 versus 84.00% for ConvNeXt). Their Top-5 accuracies were nearly identical, indicating that both architectures frequently ranked the correct genus among their highest-confidence predictions.

At the species level, ResNet50 retained performance close to its genus-level results, while both ConvNeXt variants presented a pronounced decline. The resulting Weighted F1-score gap between ResNet50 and ConvNeXt exceeded 24 percentage points. Although the two ConvNeXt species models used different initialization strategies, they performed similarly. Transfer from the genus-level model did not improve Test Accuracy and produced only a small difference in Weighted F1-score relative to training from scratch.

Table 1: Comparative performance metrics of the evaluated 1D-CNN architectures at the genus and species taxonomic levels. Results denote the evaluation on the held-out fragment-level test sets without nucleotide corruption.

Taxonomic Level	Model Architecture	Test Acc.	Top-5 Acc.	Weighted F1	Test Loss
Genus (2,702 classes)	ResNet50	86.38%	92.56%	86.19%	0.77
	ConvNeXt	85.00%	92.50%	84.00%	0.72
Species (13,683 classes)	ResNet50	84.88%	90.89%	84.58%	0.91
	ConvNeXt (from scratch)	59.18%	70.52%	59.51%	3.86
	ConvNeXt (Transfer)	58.24%	69.86%	59.87%	3.91

[**Nota v2:** (i) Os valores do ConvNeXt no nível de gênero (85,00%, 92,50%, 84,00%) parecem arredondados; reportar com duas casas decimais, como os demais. (ii) O modelo ConvNeXt_F (61,14% na Tabela 3) não consta da Tabela 1; incluí-lo ou justificar a omissão.]

3.1.1 Inference Efficiency

The computational comparison further favored ResNet50 (Table 2). ConvNeXt contained approximately 3.7 times as many parameters as ResNet50 and had a proportionally larger memory footprint, yet this additional capacity did not translate into higher classification performance. ResNet50 also had approximately 19% lower batch latency and processed roughly 23% more sequences per second.Variants sharing the same backbone had nearly identical inference measurements, as expected because their inference graphs were unchanged by the training strategy. [**Nota v2:** Informar GPU, precisão numérica (FP32 ou AMP), versão do PyTorch e se o perfil foi medido com a cabeça de espécie (13,683 classes) ou de gênero; o número de parâmetros depende fortemente da cabeça de classificação (p.ex., $2048 \times 13{,}683 \approx 28$M parâmetros apenas na camada final do ResNet50, se a dimensão latente for 2048).]

Table 2: Inference efficiency profiling of the evaluated 1D-CNN architectures.

Model	Parameters	Model Size	Batch Latency (ms)	Throughput (seq/s)
ConvNeXt_F (Noise-Aware)	164.01M	625.66 MB	4.06 ± 0.09	7,889.8
ConvNeXt_Z (Scratch)	164.01M	625.66 MB	4.06 ± 0.07	7,877.0
ConvNeXt_T (Transfer)	164.01M	625.66 MB	4.08 ± 0.05	7,844.1
ResNet50 (Standard)	43.99M	168.02 MB	3.30 ± 0.04	9,686.3
ResNet50 (Noise-Aware)	43.99M	168.02 MB	3.31 ± 0.07	9,666.2

Note: Latency was measured per inference batch of 32 sequences. Throughput is reported in sequences per second.

3.1.2 Performance Decay of Standard Clean-Trained Models

The two clean-trained architectures responded differently to increasing substitution noise (Table 3 and Fig. 1). ResNet50 produced better results on uncorrupted sequences, particularly at the species level, but this advantage disappeared after only modest sequence corruption. At $\epsilon = 0.02$, its species accuracy dropped by more than 40 percentage points, from 84.88% to 44.14%. [**Nota v2:** [Contribuição] Uma perda superior a 40 pontos com apenas 1,5% de troca efetiva de bases é compatível com um modelo que aprendeu correspondências quase exatas de k-mers, isto é, memorização dos genomas de referência. Esse resultado é informativo e merece ser discutido como tal, mas reforça a necessidade de uma partição por genoma de origem para separar generalização de memorização.] ConvNeXt started from a considerably lower clean baseline (59.18%) but was less affected by the same perturbation, retaining 51.90% accuracy.

This difference became more pronounced as the corruption rate increased. At $\epsilon = 0.05$, ConvNeXt maintained 36.86% species accuracy compared with 13.17% for ResNet50, reversing the ranking observed on uncorrupted sequences. The same

pattern was observed at the genus level: although ResNet50 performed better without corruption, ConvNeXt retained more of its predictive performance as resampling increased.

Neither clean-trained architecture remained effective under severe corruption. At $\epsilon = 0.15$, species accuracy fell below 4% for both models. ConvNeXt still remained ahead of ResNet50 under noisy conditions, although it was less accurate on uncorrupted species sequences.

3.1.3 Effects of Transfer Learning and Noise-Aware Fine-Tuning

Noise-aware fine-tuning improved robustness more effectively than genus-level transfer, although its effect differed between architectures (Table 3). For ConvNeXt, exposure to substitution noise preserved performance on uncorrupted sequences while producing a clear advantage as corruption increased: at $\epsilon = 0.15$, `ConvNeXt_F` retained 17.38% species accuracy, compared with 4.50% for the genus-transfer model (`ConvNeXt_T`) and 3.66% for the model trained from scratch (`ConvNeXt_Z`). ResNet50 showed a less favorable trade-off, as noise-aware fine-tuning improved its performance under corruption but reduced clean species accuracy from 84.88% to 32.36%. At $\epsilon = 0.10$, `ResNet50_F` retained 9.44% accuracy compared with 1.45% for the clean-trained ResNet50, but remained well below `ConvNeXt_F` under the same conditions. `ConvNeXt_F` achieved the highest accuracy across the corrupted conditions while retaining substantially more of its uncorrupted performance than noise-aware ResNet50. Because both noise-aware models were fine-tuned with $\epsilon = 0.05$, their performance at this level reflects in-distribution corruption, whereas $\epsilon \in \{0.10, 0.15\}$ represents stronger corruption than that seen during fine-tuning. [**Nota v2:** Informar quantas realizações independentes de corrupção foram avaliadas por nível de ϵ e, se mais de uma, reportar média $\pm$ desvio-padrão.]

Table 3: Single-window (512 bp) accuracy degradation under controlled nucleotide resampling.

Model	Clean (0%)	$\epsilon = 0.02$	$\epsilon = 0.05$	$\epsilon = 0.10$	$\epsilon = 0.15$
ResNet50 (Standard)	**86.38%**	49.22%	18.21%	4.08%	1.54%
ConvNeXt (Standard)	85.00%	**68.31%**	**47.32%**	**21.58%**	**9.24%**
ResNet50 (Standard)	**84.88%**	44.14%	13.17%	1.45%	0.30%
ResNet50 (Noise-Aware)	32.36%	27.08%	19.31%	9.44%	3.87%
ConvNeXt_Z (Scratch)	59.18%	51.90%	36.86%	13.99%	3.66%
ConvNeXt_T (Transfer)	58.24%	52.87%	42.02%	17.51%	4.50%
ConvNeXt_F (Noise-Aware)	61.14%	**55.34%**	**46.52%**	**31.07%**	**17.38%**

Note: The first two rows evaluate performance at the Genus level (C = 2,702 classes), while the remaining rows evaluate performance at the Species level (C = 13,683 classes).

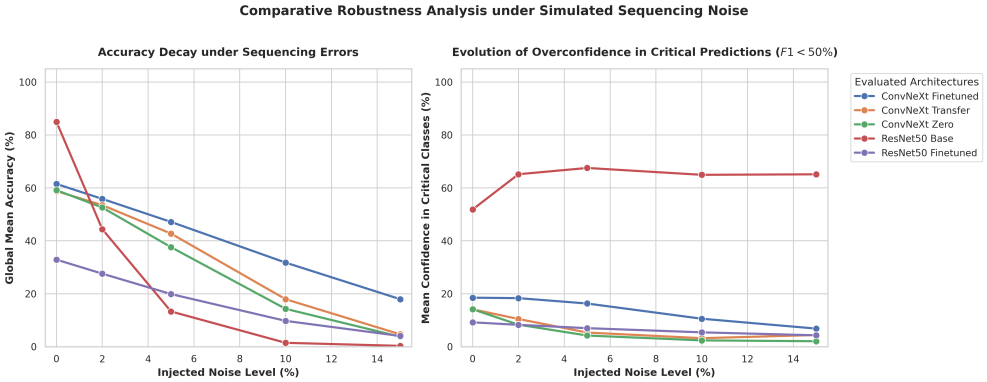


Fig. 1: Comparative robustness analysis of the evaluated 1D-CNN architectures under controlled nucleotide resampling.

3.1.4 Class-Level Confidence and Performance Analysis

Species-level performance deteriorated progressively as sequence corruption increased, although the rate of degradation differed substantially across architectures (Table 4). On uncorrupted sequences, the standard ResNet50 had the largest number of species in the Excellent performance band ($F1 \geq 0.95$), with 7,362 of 13,683 species, compared with approximately 4,000–4,400 for the ConvNeXt variants. This advantage decreased rapidly under corruption. At $\epsilon = 0.02$, only 124 species remained in the Excellent band for ResNet50, and at $\epsilon = 0.10$ all species had moved to the Poor/Critical band ($F1 < 0.50$). ConvNeXt degraded more gradually, particularly after noise-aware fine-tuning.

Standard ResNet50 remained highly confident even after its class-level predictive performance had deteriorated. [**Nota v2:** [Ref.] Três referências já presentes no `.bib` e não citadas são diretamente pertinentes à interpretação, como hipóteses e não como conclusões: `hein2019relu` (redes ReLU produzem confiança alta longe dos dados de treino; o ResNet50 usa ReLU e o ConvNeXt usa GELU), `schneider2020corruptions` (robustez a corrupções e estatísticas de Batch Normalization; o ResNet50 usa BatchNorm e o ConvNeXt usa LayerNorm) e `minderer2021revisiting` (a calibração depende da família de arquitetura). `hendrycks2019benchmarking` pode fundamentar o protocolo de corrupção controlada.] Once all species had entered the Poor/Critical band at $\epsilon = 0.10$, their mean Top-1 confidence was still 64.93%, and it remained essentially unchanged at $\epsilon = 0.15$ (65.12%). ConvNeXt had the opposite behavior: confidence decreased together with performance. For `ConvNeXt_F`, mean confidence among Poor/Critical species fell from 18.51% on uncorrupted sequences to 6.82% at $\epsilon = 0.15$, with broadly similar downward trends for the scratch and genus-transfer variants, although the latter showed a slight increase between $\epsilon = 0.10$ and $\epsilon = 0.15$ (from 3.21% to 4.41%). Noise-aware fine-tuning also changed this behavior in ResNet50: confidence among Poor/Critical species decreased from 9.20% to 4.29% as noise increased, rather than remaining high as in the standard model. The comparison separates two aspects of robustness: ResNet50 provided stronger class-level performance on uncorrupted sequences, whereas ConvNeXt preserved performance longer under corruption and its confidence more closely tracked the observed degradation in class-level F1 performance.

Table 4: Species-level performance distribution and mean predictive confidence under increasing sequence corruption. Each cell reports the number of species followed by the mean Top-1 confidence (%). Performance bands are defined according to the species-level One-vs-Rest F1-score. A dash indicates that no species were assigned to that band. [**Nota v2:** Os limites originais (0,90–0,94; 0,80–0,89; 0,50–0,79) deixavam lacunas (p.ex., F1 = 0,945); a v2 usa intervalos semiabertos. Confirmar que corresponde à implementação.]

Model	Noise	Excellent $\geq$ **0.95**	Very Good [**0.90, 0.95**)	Good [**0.80, 0.90**)	Borderline/ Fair [**0.50, 0.80**)	Poor/ Critical $<$ **0.50**
ResNet50	0%	7,362 (99.36)	742 (95.72)	869 (85.84)	3,664 (69.26)	1,046 (51.84)
	2%	124 (96.86)	464 (94.39)	1,729 (90.25)	4,777 (83.39)	6,589 (65.15)
	5%	1 (96.80)	–	9 (89.66)	789 (80.56)	12,884 (67.52)
	10%	–	–	–	–	13,683 (64.93)
	15%	–	–	–	–	13,683 (65.12)
ResNet50 noise-aware	0%	318 (68.57)	663 (60.09)	1,384 (47.50)	2,349 (29.18)	8,969 (9.20)
	2%	86 (62.27)	311 (55.33)	992 (46.16)	2,614 (26.58)	9,680 (8.25)
	5%	6 (57.37)	50 (49.04)	393 (40.00)	2,218 (22.63)	11,016 (7.00)
	10%	–	–	7 (30.19)	653 (16.37)	13,023 (5.41)
	15%	–	–	–	34 (12.97)	13,649 (4.29)
ConvNeXt scratch	0%	4,124 (59.91)	1,086 (38.09)	1,154 (29.78)	1,713 (29.81)	5,606 (14.13)
	2%	1,930 (36.60)	1,551 (28.62)	1,601 (23.43)	2,289 (17.38)	6,312 (8.31)
	5%	149 (20.30)	440 (16.62)	1,493 (12.20)	3,339 (8.28)	8,262 (4.22)
	10%	–	2 (8.54)	30 (5.24)	1,188 (2.94)	12,463 (2.37)
	15%	–	–	–	18 (1.74)	13,665 (2.06)
ConvNeXt transfer	0%	3,977 (55.79)	1,282 (38.13)	1,266 (29.85)	1,546 (31.70)	5,612 (14.12)
	2%	2,427 (36.54)	1,572 (26.81)	1,541 (21.37)	2,057 (19.81)	6,086 (10.47)
	5%	527 (18.22)	1,076 (14.65)	1,950 (11.23)	2,794 (8.70)	7,336 (5.33)
	10%	2 (4.70)	4 (6.29)	175 (4.87)	2,171 (3.75)	11,331 (3.21)
	15%	–	–	–	65 (3.56)	13,618 (4.41)
ConvNeXt noise-aware	0%	4,426 (86.17)	1,179 (63.66)	1,099 (49.34)	1,489 (45.02)	5,490 (18.51)
	2%	3,162 (78.01)	1,310 (60.59)	1,379 (44.71)	1,841 (37.27)	5,991 (18.34)
	5%	1,361 (63.67)	1,292 (49.43)	1,836 (37.24)	2,313 (27.32)	6,881 (16.31)
	10%	124 (37.08)	374 (31.37)	1,069 (24.86)	3,117 (16.18)	8,999 (10.56)
	15%	2 (16.42)	18 (16.76)	166 (13.72)	1,718 (9.91)	11,779 (6.82)

3.2 Latent Representation and Taxonomic Organization

We examined whether the latent representations learned by the 1D-CNNs reflected taxonomic organization at the order, genus, and species levels. ~~Quantitative analyses were performed directly in the original high-dimensional feature spaces.~~ We considered two complementary views of the latent geometry: Silhouette Score [24] for global class separation and k-nearest-neighbor (KNN) accuracy for local taxonomic consistency (Table 5). UMAP projections (Fig. 2) were used only for qualitative visualization.

The two metrics revealed different aspects of the learned representations. Silhouette Scores were negative for every architecture and taxonomic level, showing little evidence of globally isolated taxonomic clusters. ConvNeXt consistently produced less-negative scores than ResNet50, but even its representations remained far from a clearly separated global structure. KNN accuracy presented a different pattern. Local neighborhoods were strongly taxonomically organized, particularly for ConvNeXt, which outperformed ResNet50 at all three taxonomic levels. At the species level, for example, the ConvNeXt variants reached approximately 62–63% KNN accuracy, compared with 52–53% for ResNet50.

This local organization also followed taxonomic resolution. KNN accuracy was highest at order, decreased at genus, and was lowest at species for every model. ConvNeXt transfer, for example, decreased from 86.50% at order to 80.76% at genus and 63.01% at species. Despite this decline, local taxonomic consistency remained substantial across the 13,683 species classes, showing that finer taxonomic resolution weakened but did not eliminate neighborhood structure.

Parent–child correlations provided a second view of this hierarchy (Table 6). KNN organization was strongly preserved between adjacent taxonomic levels across all architectures. Species–genus correlations ranged from $\rho = 0.850$ to 0.883, while genus–order correlations were even stronger, ranging from $\rho = 0.918$ to 0.944. Because the parent-level KNN accuracy is computed from the same query embeddings as the child-level values, and a correct species neighborhood implies a correct genus neighborhood, part of this correspondence is expected by construction. [**Nota v2:** Os intervalos originais (0,854–0,882 e 0,917–0,943) não incluíam o valor do ConvNeXt_T (0,8496), citado logo adiante como 0,849, nem coincidiam com o arredondamento da Tabela 6. Explicar também por que apenas 2,321 dos 2,702 gêneros entram na correlação Espécie–Gênero (p.ex., gêneros com uma única espécie).] Silhouette correlations were considerably weaker, reinforcing the difference between local taxonomic organization and global cluster separation.

The comparison between architectures also shows that hierarchical consistency and absolute neighborhood quality capture different properties of the latent space. ResNet50 had stronger species–genus KNN correlations than ConvNeXt but lower KNN accuracy. Standard ResNet50, for example, reached a correlation of $\rho = 0.883$ but only 51.96% species-level KNN accuracy, while ConvNeXt transfer reached 63.01% KNN accuracy with a lower correlation of $\rho = 0.850$.

The contrast between weak global separation and relatively strong local consistency suggests that taxonomic information is distributed through local neighborhoods rather than arranged as isolated class clusters.

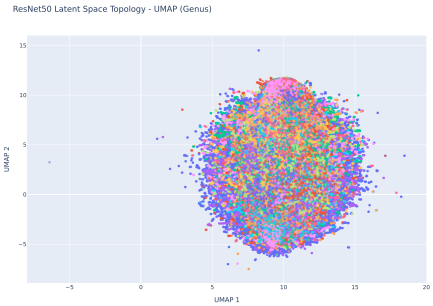
Training strategy had comparatively little effect on the ConvNeXt latent geometry. The scratch, transfer, and noise-aware variants produced similar KNN accuracies across taxonomic levels and similarly strong parent–child correlations. The larger difference was between the ResNet50 and ConvNeXt configurations.

3.3 Hierarchical Entropy Analysis

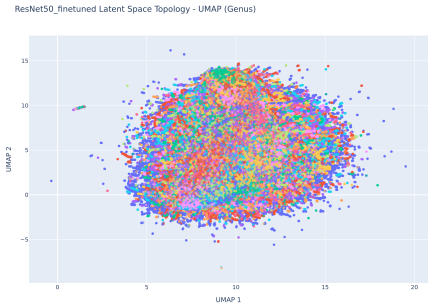
[**Nota v2:** [Coerência] Subseção movida na v2 para junto da análise do espaço latente, pois trata de incerteza preditiva e estrutura de ativação, conforme o item (iii) dos objetivos no Background.]

Entropy analysis revealed a clear separation between the standard ResNet50 and the remaining models (Table 7). [**Nota v2:** Informar a base do logaritmo (nats ou bits). Para referência, a entropia máxima é $\ln 13{,}683 \approx 9{,}52$ nats no nível de espécie.] The ConvNeXt variants and the noise-aware ResNet50 produced strong reductions in predictive uncertainty after aggregation from the species to the genus level. On the other hand, the standard ResNet50 exhibited only a small hierarchical entropy gap. This suggests that, in the former models, uncertainty was more frequently distributed among species within the same genus, and that the standard ResNet50 produced a sharper species-level predictive distribution.

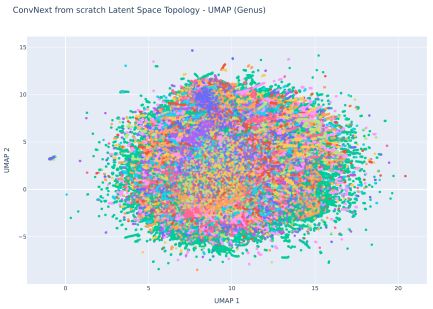
Activation entropy provided a complementary view of architectural differences. The three ConvNeXt variants exhibited similarly high normalized activation entropy,



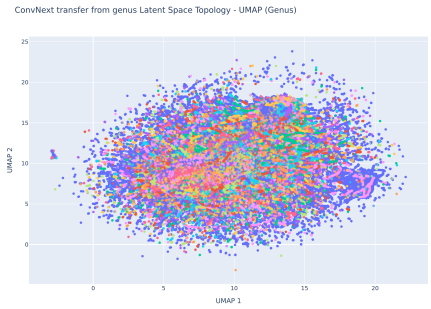
(a) ResNet50 (Standard)



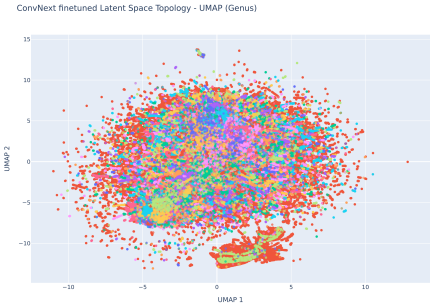
(b) ResNet50 (Noise-aware)



(c) ConvNeXt (from scratch)



(d) ConvNeXt (Transfer)



(e) ConvNeXt (Noise-aware)

Fig. 2: UMAP projections of the latent representations at the genus level across the evaluated 1D-CNN architectures.

Table 5: Global latent-space organization across taxonomic levels. Silhouette Score measures global cluster separation, while KNN Accuracy evaluates local taxonomic consistency.

Taxonomic Level	Model	Silhouette Score	KNN Accuracy (%)
Species	ConvNeXt_F (Noise-Aware)	-0.0376	62.85
	ConvNeXt_T (Transfer)	-0.0366	63.01
	ConvNeXt_Z (Scratch)	-0.0495	62.71
	ResNet50 (Standard)	-0.1882	51.96
	ResNet50_F (Noise-Aware)	-0.1887	53.49
Genus	ConvNeXt_F (Noise-Aware)	-0.0728	80.88
	ConvNeXt_T (Transfer)	-0.0739	80.76
	ConvNeXt_Z (Scratch)	-0.0781	79.09
	ResNet50 (Standard)	-0.1813	67.98
	ResNet50_F (Noise-Aware)	-0.1962	69.78
Order	ConvNeXt_F (Noise-Aware)	-0.0448	86.49
	ConvNeXt_T (Transfer)	-0.0483	86.50
	ConvNeXt_Z (Scratch)	-0.0459	84.47
	ResNet50 (Standard)	-0.0855	73.79
	ResNet50_F (Noise-Aware)	-0.0936	76.53

Table 6: Hierarchical correspondence of latent-space organization across adjacent taxonomic levels. Values report Spearman's rank correlation coefficient (ρ) for KNN Accuracy (species–genus $N = 2321$; genus–order $N = 74$) and Silhouette Score (Species–genus $N = 2154$–2166; genus–order $N = 73$–74). [**Nota v2:** A Seção de Métodos afirma que também foi calculado o r de Pearson, mas a tabela traz apenas ρ; incluir ou remover a menção.]

	KNN Accuracy (ρ)		Silhouette Score (ρ)	
Model	**Species–Genus**	**Genus–Order**	**Species–Genus**	**Genus–Order**
ConvNeXt_F (Noise-Aware)	0.8541	0.9410	0.3750	0.3767
ConvNeXt_T (Transfer)	0.8496	0.9376	0.3770	0.3322
ConvNeXt_Z (Scratch)	0.8649	0.9439	0.2610	0.3911
ResNet50 (Standard)	0.8828	0.9352	0.1520	0.4215
ResNet50_F (Noise-Aware)	0.8743	0.9178	0.2730	0.3389

suggesting broadly distributed use of the pre-classifier latent dimensions despite their different training strategies. In contrast, standard ResNet50 presented a more concentrated activation profile and noise-aware ResNet50 occupied an intermediate position. ~~The similarity among the ConvNeXt variants indicates that this aspect of latent activation structure was influenced more strongly by architecture than by the training regime.~~ The similarity among the ConvNeXt variants indicates that, within ConvNeXt, this aspect of latent activation structure was largely insensitive to the training regime;

Table 7: Hierarchical predictive entropy and normalized activation entropy across the evaluated models. ΔH denotes the species–genus hierarchical entropy gap.

Model	$\mathbf{H}_{\text{species}}$	$\mathbf{H}_{\text{genus}}$	$\mathbf{H}_{\text{activation}}$	$\mathbf{\Delta H}$
ResNet50 (Standard)	0.5264	0.4142	0.7443	0.1122
ResNet50 (Noise-Aware)	6.0811	3.8378	0.8656	2.2432
ConvNeXt_Z (Scratch)	5.6708	3.3941	0.9582	2.2767
ConvNeXt_T (Transfer)	5.8222	3.3866	0.9618	2.4356
ConvNeXt_F (Noise-Aware)	3.9058	2.1157	0.9628	1.7901

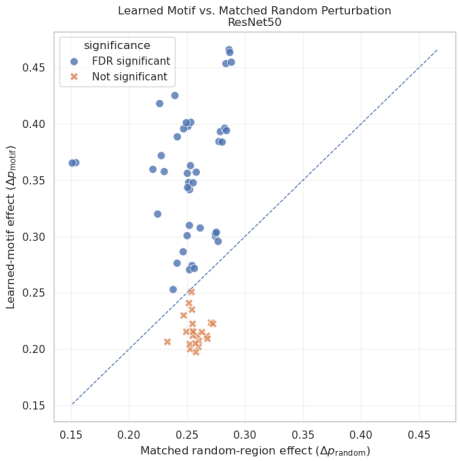
whether the difference relative to ResNet50 reflects the architecture or other aspects of the training configuration cannot be determined from the present design.

Overall, predictive uncertainty and latent activation structure varied systematically across architectures. ConvNeXt combined broader species-level uncertainty with highly distributed latent activations, whereas standard ResNet50 combined sharper predictions with a more concentrated internal representation. Fine-tuning notably altered the ResNet50 entropy profile, but produced comparatively smaller changes within ConvNeXt, further suggesting that the two architectures respond differently to the training strategy.

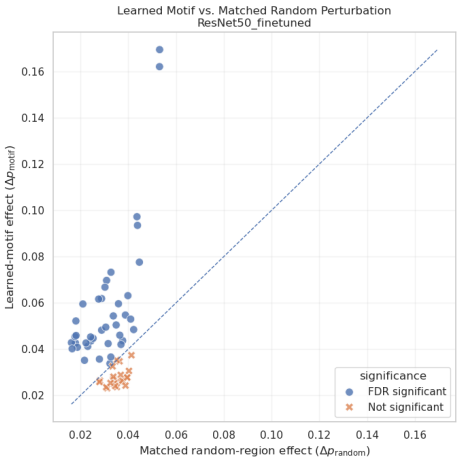
3.4 Learned Sequence Motifs

The matched perturbation analysis showed that the effects associated with the learned motifs could be distinguished from the general sensitivity of the models to local nucleotide substitutions (Fig. 3 and Table 8). For the statistically validated motifs, perturbation of the learned sequence pattern produced a larger reduction in true-class probability than perturbation of a matched random 7-bp region from the same sequence ($\Delta p_{\text{motif}} > \Delta p_{\text{random}}$). ~~Accordingly, these motifs exhibited positive motif-specific effects, $\Delta p_{\text{specific}} = \Delta p_{\text{motif}} - \Delta p_{\text{random}} > 0$, with the paired difference remaining significant after BH-FDR correction.~~ Using the validation criteria defined in Section 2.9.1~~predefined validation criteria~~, 34 motifs were retained for the noise-aware ConvNeXt, 30 for ConvNeXt trained from scratch, 40 for the genus-transfer ConvNeXt, 42 for the standard ResNet50, and 40 for the noise-aware ResNet50. ~~These results show that the predictive effect of the validated motifs exceeded the perturbation response expected from disrupting an arbitrary region of equivalent length within the same sequence.~~

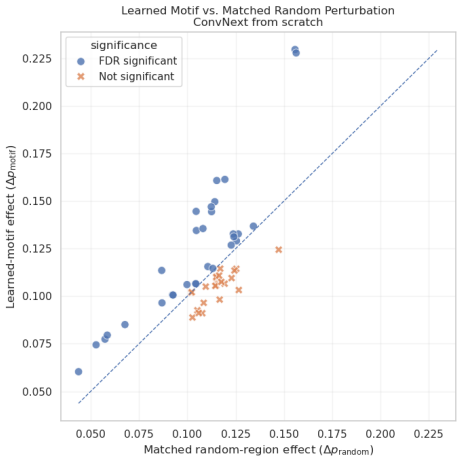
Ranking the validated motifs by their mean motif-specific probability drop ($\overline{\Delta p}_{\text{specific}}$) revealed substantial differences in effect magnitude across architectures. The largest effects were observed for the standard ResNet50, where *GGGGGGG* produced the highest motif-specific probability drop ($\overline{\Delta p}_{\text{specific}} = 0.2140$), followed by *CCCCCCC* (0.2113), *GGGGGCG* (0.1915), *GGGGCGG* (0.1855), and *AAATTTA* (0.1796). Several additional ResNet50 motifs also showed comparatively large specific effects, including *TTTAAAA* (0.1765), *TAAATAT* (0.1699), and *TTATATT* (0.1665). The noise-aware ResNet50 exhibited lower maximum effects, with *CCCCCCC* and



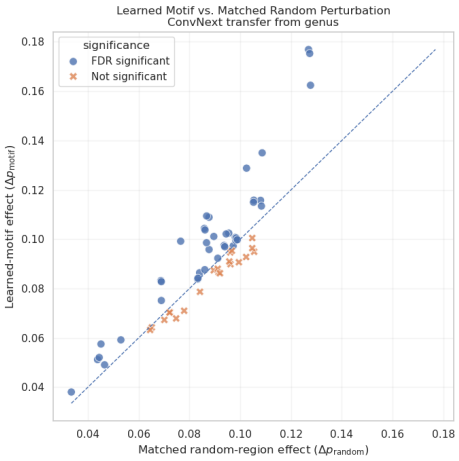
(a) ResNet50 (Standard)



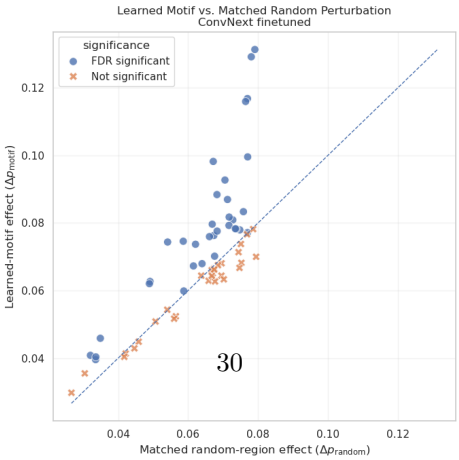
(b) ResNet50 (Noise-aware)



(c) ConvNeXt from scratch



(d) ConvNeXt (Transfer)



(e) ConvNeXt (Noise-aware)

Fig. 3: Learned-motif perturbation versus matched random-region perturbation across the evaluated architectures. Each point represents a first-layer consensus motif. Points above the identity line indicate a larger reduction in true-class probability after motif perturbation than after perturbation of the matched random control region. **[Nota v2:** Corrigido na v2: o painel (e), ConvNeXt noise-aware, exibia a imagem do ResNet50 noise-aware; e os nomes de dois arquivos ("ConvNext", com x minúsculo) não coincidiam com o `.tex` ("ConvNeXt"), o que impede a compilação em sistemas sensíveis a maiúsculas (Linux). Sugiro renomear as figuras sem espaços e corrigir o nome da pasta "mottif_plots".]

GGGGGGG reaching $\overline{\Delta p}_{\text{specific}} = 0.1163$ and 0.1090, respectively. The ConvNeXt variants showed smaller motif-specific effects overall, although homopolymeric and compositionally biased patterns such as *CCCCCCC* and *GGGGGGG* also ranked among their strongest validated motifs. Thus, the architectures differed not only in the motifs retained by their first-layer filters, but also in the magnitude with which individual sequence patterns contributed specifically to the predicted taxonomic class. taxonomic class. [**Nota v2:** [Contribuição] Os motivos com maior efeito são homopolímeros e padrões de baixa complexidade (GGGGGGG, CCCCCCC). Para distinguir sinal biológico de viés composicional, recomenda-se um controle com regiões aleatórias pareadas por composição (mesmo conteúdo GC ou mesma complexidade) em vez de regiões aleatórias quaisquer. Considerar também encurtar esta seção e a de RSCU, ou movê-las para material suplementar, já que sustentam conclusões fracas.]

Table 8: Summary of statistically validated first-layer motifs and motif-specific perturbation effects.

Model	Extracted motifs	Validated motifs	Strongest motif	Top specific effect	Motifs for 75% effect
ConvNeXt noise-aware	63	34	CCCCCCC	0.0523	15
ConvNeXt from scratch	54	30	CCCCCCC	0.0739	12
ConvNeXt transfer from genus	61	40	CCCCCCC	0.0500	15
ResNet50	64	42	GGGGGGG	0.2140	22
ResNet50 noise-aware	62	40	CCCCCCC	0.1163	20

3.4.1 Triplet-Composition Association of Learned Motifs

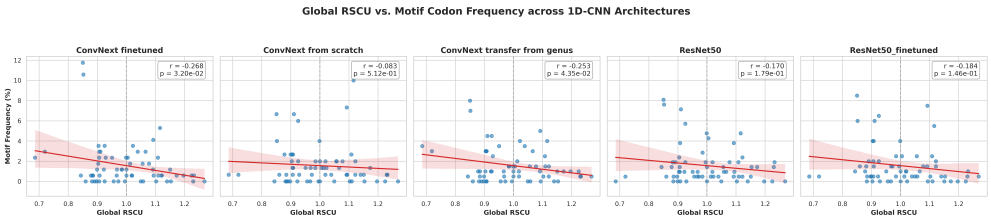


Fig. 4: Association between the RSCU-normalized background triplet profile and triplet frequency within statistically validated motifs. Each point represents one of the 64 nucleotide triplets, with global Relative Synonymous Codon Usage (RSCU) on the x-axis and its frequency among triplets extracted from model-specific 7-mer motifs on the y-axis. Solid lines indicate linear regression fits with 95% confidence intervals, and the dashed vertical line marks the neutral RSCU reference (RSCU=1). Pearson correlation coefficients (r) and corresponding p-values are reported for each model.

Table 9: Correlation between the RSCU-normalized background profile and triplet frequency within statistically validated motifs.

Model	Pearson r	p-value	Spearman ρ	p-value
ConvNeXt_F (Noise-Aware)	**-0.2684**	**0.0320**	-0.1729	0.172
ConvNeXt_Z (Scratch)	-0.0835	0.512	-0.1133	0.373
ConvNeXt_T (Transfer)	**-0.2532**	**0.0435**	-0.0899	0.480
ResNet50 (Standard)	-0.1702	0.179	-0.1002	0.431
ResNet50_F (Noise-Aware)	-0.1838	0.146	-0.2071	0.101

Triplet frequencies within the statistically validated motifs showed weak negative associations with global RSCU in all architectures (Figure 4 and Table 9). The strongest correlations were observed for noise-aware ConvNeXt ($r = -0.268$, $p = 0.032$) and genus-transfer ConvNeXt ($r = -0.253$, $p = 0.044$), while the associations were weaker and non-significant for standard ResNet50 ($r = -0.170$), noise-aware ResNet50 ($r = -0.184$), and ConvNeXt trained from scratch ($r = -0.084$). None of the Spearman correlations was statistically significant, and the two nominally significant Pearson correlations would not remain significant after correction for the ten tests performed. Overall, triplet composition did not closely follow the global usage profile. [**Nota v2:** Os valores-p não foram corrigidos para múltiplos testes (5 modelos × 2 coeficientes); com Bonferroni ou BH nenhum resultado permanece significativo. Além disso, o perfil de fundo usa tripletos não sobrepostos no quadro 0, enquanto os motivos usam tripletos sobrepostos; explicar ou uniformizar.]

Canonical translation signals were relatively uncommon among the validated motifs (Table 10). ATG-containing motifs were observed only in noise-aware ConvNeXt (2.9%) and genus-transfer ConvNeXt (2.5%), while none were detected in ConvNeXt trained from scratch or either ResNet50 configuration. Motifs containing TAA, TAG, or TGA triplets were more frequent, particularly in genus-transfer (27.5%) and noise-aware ConvNeXt (23.5%). Although the two ConvNeXt models containing ATG motifs were also the most robust ConvNeXt variants at $\epsilon = 0.15$, this pattern was not consistent across architectures: noise-aware ResNet50 retained 11.97% of its clean performance despite containing no ATG-bearing motifs.

Table 10: Prevalence of ATG and stop-associated triplets among statistically validated motifs and robustness under sequence corruption. [**Nota v2:** Cabeçalhos "START"/"STOP" renomeados para ATG e TAA/TAG/TGA, pois o texto afirma que não se trata de sinais de tradução. A acurácia retida do ConvNeXt_Z era 6,25%, mas $3{,}66/59{,}18 = 6{,}18\%$; as demais linhas conferem.]

Model	ATG (%)	TAA/TAG/TGA (%)	Acc. 0%	Acc. 15%	Retained Acc. (%)
ResNet50 (Standard)	0.0	9.5	84.88	0.30	0.35
ResNet50 (Noise-Aware)	0.0	15.0	32.36	3.87	11.97
ConvNeXt_Z (Scratch)	0.0	13.3	59.18	3.66	6.18
ConvNeXt_T (Transfer)	2.5	27.5	58.24	4.50	7.73
ConvNeXt_F (Noise-Aware)	2.9	23.5	61.14	17.38	28.42

3.4.2 Motif Co-occurrence and Triplet-Composition Similarity

Table 11: Pearson correlation between mean RSCU values of the 200 strongest co-occurring validated motif pairs.

Model	Pearson r (Top-200)
ResNet50 (Standard)	**0.827**
ResNet50 (Noise-Aware)	**0.744**
ConvNeXt_F (Noise-Aware)	0.595
ConvNeXt_T (Transfer)	0.580
ConvNeXt_Z (Scratch)	0.360

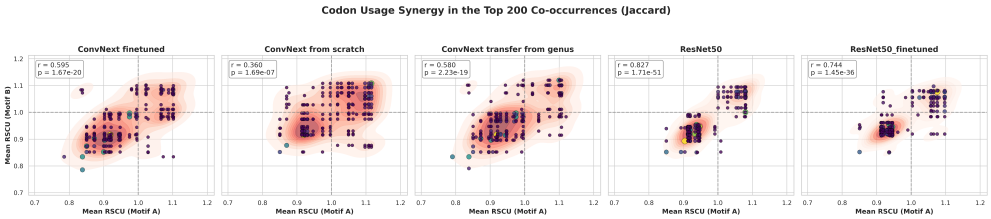


Fig. 5: RSCU-derived triplet-composition similarity among strongly co-occurring motif pairs across the evaluated 1D-CNN architectures. Each point represents one of the 200 strongest motif co-occurrences ranked by the Jaccard index, with the mean RSCU of the two paired motifs shown on the x- and y-axes. Dashed lines indicate the neutral RSCU reference (RSCU=1), while density contours show the distribution of motif pairs in RSCU space. Pearson correlation coefficients (r) and corresponding p-values quantify the association between the RSCU-derived triplet-composition profiles of co-occurring motifs.

The 200 strongest co-occurring validated motif pairs showed positive associations between their mean RSCU values across all architectures (Table 11). [Nota v2: (i) O texto afirma significância estatística, mas a Tabela 11 não traz valores-p; incluí-los. (ii) Os 200 pares não são independentes (o mesmo motivo aparece em vários pares) e o r de Pearson depende da ordem arbitrária (i, j) de cada par; recomenda-se um teste de permutação que embaralhe os pareamentos. (iii) Motivos que coocorrem tendem a ter composição semelhante (p.ex., GGGGGGG e GGGGGCG em regiões ricas em GC), o que torna a correlação parcialmente esperada; discutir esse viés.] The strongest relationship was observed for standard ResNet50 ($r = 0.827$), followed by noise-aware ResNet50 ($r = 0.744$). The ConvNeXt models presented weaker but still clear associations, with fine-tuned and genus-transfer variants reaching $r = 0.595$ and $r = 0.580$, respectively, while the scratch model had the lowest correlation ($r = 0.360$) (Figure 5).

Strongly co-occurring motif pairs occurred on both sides of the neutral $RSCU = 1$ reference, showing that this similarity was not restricted to motifs associated with

preferentially used triplets. Together with the single-motif analysis, these results indicate that individual motif composition was only weakly related to global triplet usage, whereas motifs that frequently co-occurred exhibited ~~considerably~~ stronger similarity in their triplet-usage profiles. This similarity, however, is expected in part from the shared nucleotide composition of co-occurring motifs and was not tested against a null model; it should therefore be regarded as exploratory.

3.5 Long-Sequence Classification via Sliding-Window Consensus

Table 12: Performance of sliding-window consensus for long-sequence classification. Results are reported at the species and genus levels using majority voting and mean-logit aggregation; species-level Top-5 accuracy is also shown.

	Species (%)			Genus (%)	
Model	**Majority**	**Mean Logits**	**Top-5**	**Majority**	**Mean Logits**
ResNet50	**66.67**	**72.69**	**82.51**	**75.16**	**81.29**
ResNet50 noise-aware	22.10	21.54	30.23	23.65	22.94
ConvNeXt noise-aware	20.38	19.99	34.13	22.06	21.55
ConvNeXt scratch	12.01	12.00	23.09	12.35	12.32
ConvNeXt transfer	3.49	3.71	13.75	4.68	4.90

~~Sliding-window consensus allowed all five models trained on 512 bp fragments to classify variable-length sequences, but~~ When the sliding-window consensus was applied to the long-sequence dataset, species-level performance differed across architectures and training strategies. [**Nota v2:** (i) O desempenho em sequências longas dos modelos ConvNeXt (3,49–20,38%) é muito inferior ao obtido nos fragmentos (58–61%), e o ConvNeXt_T cai para 3,49% enquanto o ConvNeXt_Z obtém 12,01%. Uma queda dessa magnitude, ausente no ResNet50 (84,88% → 72,69%), merece investigação (p.ex., mapeamento de rótulos, tratamento de bases IUPAC ou de janelas finais) e explicação explícita. (ii) As sequências completas foram obtidas do NCBI para espécies presentes no treinamento e podem coincidir com os genomas de referência dos quais os fragmentos de treino foram derivados. Quantificar essa sobreposição, pois ela determina se o experimento mede generalização.] ResNet50 achieved the highest mean-logit accuracy by a large margin, reaching 72.69% (Table 12). The remaining models performed considerably worse. Within the ConvNeXt family, fine-tuning produced a clear improvement over the model trained from scratch. Mean-logit Top-1 accuracy increased from 12.00% to 19.99%, corresponding to a relative improvement of approximately 66.6%.

As shown in Table 12, neither consensus strategy consistently outperformed the other at Top-1. The larger Top-5 values, particularly for noise-aware ConvNeXt, show that the correct species often remained among the highest-ranked predictions even when it was not selected as the final class.

3.5.1 Class-Level Performance and Confidence on Long Sequences

The advantage of ResNet50 on clean long sequences was also evident at the class level. ResNet50 placed 7,600 species in the Excellent band ($F1 \geq 0.95$), compared with 4,553 for noise-aware ConvNeXt. A further 774, 1,037, and 1,762 ResNet50 classes were classified as Very Good, Good, and Borderline/Fair, respectively, leaving 1,860 species in the Poor/Critical band. For noise-aware ConvNeXt, the corresponding counts were 268, 447, and 780, while 6,985 species remained below $F1 = 0.50$.

The confidence associated with these classes, however, showed a different pattern. ResNet50 assigned a mean confidence of 96.36% to species in the Excellent band, but confidence remained as high as 80.70% among species in the Poor/Critical band. Fine-tuned ConvNeXt presented a much larger separation: mean confidence decreased from 76.81% for Excellent species to 30.91% for Poor/Critical species. Thus, although ResNet50 classified more species correctly on clean long sequences, its confidence remained high even for classes with poor F1-scores.

3.5.2 Robustness of Long-Sequence Consensus under Sequence Corruption

Sequence corruption substantially changed this picture. At $\epsilon = 0.05$, only 46 ResNet50 species remained in the Excellent band, while 11,538 fell below $F1 = 0.50$, but the Poor/Critical group still had a mean confidence of 80.33%, only slightly below the value observed for poorly performing classes on clean sequences.

Fine-tuned ConvNeXt was considerably more resilient at the same noise level. It retained 1,442 species in the Excellent band, while 9,732 fell into the Poor/Critical category. Its confidence also responded more clearly to the loss in predictive performance, decreasing from 47.72% among Excellent species to 32.94% among Poor/Critical species.

At $\epsilon = 0.10$, performance deteriorated sharply for both architectures. Fine-tuned ConvNeXt retained only 26 species in the Excellent band, with 12,599 species classified as Poor/Critical. ResNet50 deteriorated even further, with only one species remaining in the Borderline/Fair category and nearly the entire evaluated taxonomic space falling into the lowest performance band.

ResNet50 was stronger on clean long sequences, but this advantage disappeared rapidly as nucleotide substitutions increased, while its confidence remained high. Fine-tuned ConvNeXt started from a lower clean-sequence accuracy but retained useful predictions for more species under moderate corruption and reduced its confidence as performance declined.

3.5.3 Per-Species Differences Between Architectures

The per-species comparison further shows that neither architecture dominated uniformly once sequence noise was introduced. At $\epsilon = 0.05$, ResNet50 produced more correct predictions than noise-aware ConvNeXt for 4,807 species, while ConvNeXt performed better for 4,068. Another 186 species had equal non-zero numbers of correct predictions, while both models failed completely for 3,972 species.

At $\epsilon = 0.10$, the balance among the remaining classifiable species shifted toward noise-aware ConvNeXt. It outperformed ResNet50 for 1,368 species, compared with 1,202 species for which ResNet50 performed better. [**Nota v2:** Em $\epsilon = 0{,}10$ as categorias somam 13,028 das 13,033 espécies; reportar os 5 empates restantes, como feito para $\epsilon = 0{,}05$.] At this level of corruption, however, both architectures failed completely for 10,458 species, showing that only a relatively small fraction of the taxonomic space remained distinguishable.

Some species showed particularly large model-specific differences. At $\epsilon = 0.10$, noise-aware ConvNeXt correctly classified 115 of 131 samples of *Biavirus raunefjordenense* [**Nota v2:** Definir "sample" neste contexto (sequências distintas, realizações de ruído ou janelas), já que a classificação é feita por sequência.], compared with only 16 for ResNet50. Similar differences were observed for *Theiavirus salishense* (87 versus 17 correct predictions) and *Cotonvirus japonicum* (66 versus 8).

~~Sliding-window aggregation allowed classifiers trained on 512-bp fragments to generate sequence-level predictions for the variable-length genomic sequences in the long-sequence dataset.~~ In the ConvNeXt predictions, several unrelated species were repeatedly assigned to taxa such as *Bracoviriform facetosae* and *Pandoravirus dulcis*. A similar pattern appeared after aggregation to the genus level, where multiple true genera were repeatedly assigned to *Bracoviriform*. These recurrent assignments indicate the presence of prediction "sinks" that attracted errors from multiple parts of the taxonomy.

The sliding-window approach enabled the application of fixed-length models to variable-length sequences, while revealing a clear divergence between the strengths of the ResNet50 and ConvNeXt configurations. On uncorrupted data, ResNet50 achieved higher accuracy. However, when faced with sequence noise, noise-aware ConvNeXt sustained accuracy across more species, while its confidence decreased more consistently as predictive performance deteriorated. Neither model emerges as a universal solution under degradation; the per-species differences suggest taxon-dependent complementarity, although the present analysis does not establish systematic clade-specific specialization (Section 4). [**Nota v2:** O original afirmava "highly specific advantages for different taxonomic clades", o que contradiz a Discussão ("not sufficient to establish systematic clade-specific specialization"). O título da subseção foi ajustado pelo mesmo motivo.]

4 Discussion

Base-level one-hot representations of viral genomic DNA supported high-cardinality taxonomic classification. However, predictive accuracy alone provides an incomplete picture. The evaluated ResNet50 and ConvNeXt configurations differed in clean-sequence accuracy, degradation under nucleotide resampling, confidence behavior, and latent-space organization. These properties describe complementary dimensions of model behavior.

The clearest distinction emerged between clean-sequence performance and perturbation robustness. Standard ResNet50 established a strong species-level baseline on uncorrupted sequences. However, this advantage vanished rapidly under controlled nucleotide resampling. ConvNeXt degraded more gradually. Noise-aware fine-tuning amplified this difference, improving ConvNeXt's robustness while

severely reducing ResNet50's clean-sequence accuracy. ~~This asymmetry suggests that learning discriminative features from uncorrupted data and maintaining informative representations under noise are distinct optimization objectives.~~ Under the fine-tuning protocol used here, these two objectives conflicted for ResNet50 but were compatible for ConvNeXt, whose clean-sequence accuracy increased from 58.24% to 61.14% after noise-aware fine-tuning. Because both models were fine-tuned with the same learning rate and number of epochs, but from different starting checkpoints, this asymmetry may reflect the fine-tuning protocol as well as intrinsic architectural properties. [**Nota v2:** [Contribuição] Para que as comparações entre arquiteturas sejam estatisticamente defensáveis, reportar intervalos de confiança por *bootstrap* sobre o conjunto de teste para todas as acurácias e F1 e, idealmente, repetir o treinamento dos dois melhores modelos com sementes diferentes.]

The models also differed in predictive confidence. ConvNeXt's confidence scaled more consistently with performance degradation. This comparison, however, is confounded by the training schedules: label smoothing ($\alpha = 0.1$), which reduces predictive confidence, was applied during species-level training of the ConvNeXt models but not of the standard ResNet50. Moreover, the ConvNeXt models trained from scratch and by genus transfer were underconfident even for well-classified species, with a mean confidence of 56–60% in the Excellent band (Table 4). Confidence that decreases with performance should therefore not be equated with good calibration. Modern neural networks frequently produce miscalibrated confidence estimates, particularly under distribution shift [26, 27], and our use of descriptive Top-1 confidence precludes formal calibration claims. Nevertheless, for applications with variable sequence quality, this divergence in confidence response remains a practical factor for model selection. [**Nota v2:** [Contribuição] Como o comportamento da confiança é um argumento central do artigo, recomenda-se calcular métricas formais de calibração (ECE, Brier *score*, NLL) e diagramas de confiabilidade para cada modelo e nível de ϵ, em vez de deixá-las como trabalho futuro. O `guo2017calibration`, já citado, fornece a metodologia.]

The strong fine-grained classification performance observed here also contrasts with the results previously obtained on the underlying taxonomic dataset using genomic foundation models. Schoier [15] evaluated eight pretrained DNA Transformer backbones on the same 2,702-genus and 13,683-species classification problems. Despite large-scale pretraining, these models plateaued at approximately 0.15–0.20 Macro-F1 at the genus level and typically 0.05–0.10 at the species level.

Although Schoier [15] reported Macro F1 whereas Weighted F1 is used here, the two metrics coincide when all classes have equal support; because both studies are based on class-balanced versions of the underlying dataset, they are expected to be close, although not necessarily identical if the per-class test support deviates from perfect balance. [**Nota v2:** Como o Macro F1 é trivial de calcular a partir das predições existentes, recomenda-se reportá-lo diretamente, o que elimina essa ressalva.] Nevertheless, the independently generated train and evaluation partitions prevent a controlled head-to-head comparison.

Despite this limitation, the magnitude of the difference is noteworthy. Schoier reported severe performance ceilings for genomic foundation models at the finest taxonomic ranks, with species-level Macro-F1 typically around 0.05–0.10. In contrast, the convolutional models evaluated here achieved substantially higher fine-grained classification performance. This difference should not be interpreted as proof of architectural superiority because a direct and controlled comparison remains necessary.

The latent-space analyses likewise showed that global and local organization captured different properties of the learned representations. Silhouette Scores remained negative across all taxonomic ranks, indicating limited global separation into isolated class clusters. In contrast, KNN accuracy was substantially above the chance level expected for a high-cardinality label space and increased systematically from species to genus and order. ConvNeXt produced higher KNN accuracy than ResNet50 at every evaluated rank. This divergence illustrates why genomic embedding quality should not be summarized using a single geometric measure [25]. **[Nota v2:** [Coerência] Há uma inversão não discutida: no ConvNeXt, a acurácia KNN de espécie (≈63%) supera a do próprio classificador (58–61%), enquanto no ResNet50 a acurácia KNN (≈52%) é muito inferior à do classificador (84,88%). Isso sugere que, no ConvNeXt, o gargalo está na camada de classificação e não na representação, e que o ResNet50 classifica bem apesar de vizinhanças locais menos organizadas. Recomenda-se discutir explicitamente essa inversão; uma sonda linear (*linear probe*) ou um KNN como classificador alternativo ajudariam a testá-la.]

The coexistence of weak global separation and comparatively high local taxonomic consistency is consistent with representations in which taxonomic information is organized primarily through local neighborhood relationships rather than globally isolated clusters. Closely related viral taxa can share substantial sequence similarity [19, 28] **[Nota v2:** [Ref.] `simmonds2017virus` sustenta a afirmação; `lefkowitz2018virus` descreve o banco de dados do ICTV e é mais adequado nos Métodos (onde foi incluído na v2) do que como evidência de similaridade de sequência.], making such a graded organization biologically plausible. Related representation-learning approaches have also shown that sequence embeddings can retain taxonomic, hierarchical, or evolutionary information [29–31]. **[Nota v2:** [Ref.] `deephier2021` trata de *embeddings* hierárquicos de sequências proteicas (GPCRs), não genômicas. Como a frase fala de *embeddings* de sequência em geral e a seguinte destaca que esses métodos incorporam informação hierárquica no objetivo, a citação é aceitável; convém apenas deixar explícito que se trata de proteínas.] Importantly, several of these methods explicitly incorporate hierarchical, metric, or phylogenetic information into their learning objectives. No such constraint was imposed here.

Parent–child correlations show consistency across adjacent ranks, particularly for the KNN-based measure, although part of this correspondence is expected by construction, because parent- and child-level metrics are computed from the same query embeddings. This should not be interpreted as reconstruction of viral phylogeny. Genus and order assignments were derived from the curated taxonomy, and the models were optimized for supervised taxonomic classification rather than evolutionary-distance recovery. Establishing a phylogenetic interpretation would require direct comparison between latent and reference phylogenetic distances.

Matched perturbation provided a complementary view of the local sequence information used by the classifiers. For a subset of first-layer filter-derived patterns, perturbing a matched occurrence in a held-out sequence reduced true-class probability more strongly than perturbing a random region of equal length from the same sequence using the same nucleotide substitutions. This supports pattern-specific predictive sensitivity beyond generic local perturbation. Nevertheless, first-layer convolutional filters should not automatically be treated as complete biological motifs. Previous analyses of genomic CNNs have shown that relevant sequence features can be distributed across multiple filters or layers [18]. Moreover, because the consensus motifs correspond to the sequences that maximally activate first-layer filters, perturbing their occurrences

may affect downstream activations more than perturbing random regions, independently of any biological meaning; the matched control removes the generic effect of substitution but not this selection effect. The recurrent homopolymeric and compositionally biased patterns observed here may therefore reflect biologically informative sequence composition, low-complexity sequence features, database specific signals, or a combination of these factors.

The downstream nucleotide-triplet analysis provides only exploratory evidence about the composition of these validated patterns. Their triplet frequencies showed weak associations with the RSCU-normalized background profile~~, with statistical support varying across model configurations~~, none of which remained statistically significant after correction for multiple testing. Because the evaluated fragments were not restricted to annotated coding sequences and were not aligned to biological reading frames, these analyses do not directly measure translational codon usage. Similarly, occurrences of ATG, TAA, TAG, or TGA within 7-mer patterns should be interpreted as nucleotide-triplet composition rather than direct evidence for translation initiation or termination signals. Future coding-aware analyses should control for reading frame, coding status, GC/GC3 content, dinucleotide composition, and amino-acid composition.

The entropy analysis provides another perspective on model uncertainty. Because genus is deterministically defined by species in the curated taxonomy, the difference $H_{\text{species}} - H_{\text{genus}}$ corresponds to the conditional entropy $H(S \mid G)$ of the predicted distribution, averaged over test sequences. The larger values observed for ConvNeXt and noise-aware ResNet50 therefore indicate greater residual uncertainty among species belonging to the same genus. Standard ResNet50 instead produced a much sharper species-level distribution. Its low predictive entropy should not be equated with reliable uncertainty, particularly given the high confidence retained under sequence corruption.

Sliding-window aggregation showed that the fixed-length classifiers can be applied to longer contiguous genomic sequences without modifying their underlying architecture. Majority voting and mean-logit aggregation produced similar Top-1 results for several configurations, while standard ResNet50 achieved the highest clean long-sequence performance. Fragment-aggregation strategies are also used in metagenomic classifiers such as MetageNN [12]. **[Nota v2:** [Ref.] Confirmar no artigo do MetageNN que ele classifica *long reads* agregando predições de fragmentos; caso contrário, reformular a frase.] The present long-sequence dataset, however, consists of continuous genomic reference sequences rather than experimental error-prone long reads. These results therefore establish the feasibility of variable-length inference but not robustness to the full error characteristics of Oxford Nanopore sequencing. Moreover, because the long sequences were obtained for species represented in training, they may overlap with the reference genomes from which the training fragments were derived; the long-sequence results should therefore not be interpreted as generalization to unseen genomes.

Per-species comparisons further showed that model performance under corruption was heterogeneous across the taxonomic label space. At $\epsilon = 0.05$, standard ResNet50 remained superior for more species, whereas at $\epsilon = 0.10$ noise-aware ConvNeXt performed better for more of the remaining distinguishable species. Recurrent prediction sinks also showed that errors under corruption were not distributed uniformly among

output classes. These results motivate future investigation of taxon-dependent model complementarity, but the present analysis is not sufficient to establish systematic clade-specific specialization.

[**Nota v2:** Removido na v2 um parágrafo que repetia a comparação com Schoier (2025), já feita em três parágrafos anteriores. A frase final ("A direct comparison would require ... same sequence partitions, preprocessing protocol, optimization budget, and corruption conditions") foi incorporada às Limitações.]

4.1 Limitations

Several limitations shape the interpretation of these findings and motivate future research. First, our experiments serve as a controlled stress test for substitution noise, but they do not capture the full complexity of real Oxford Nanopore (ONT) errors, such as indels and homopolymer errors [6–8, 32]. [**Nota v2:** [Ref.] No `.bib`, `huang2021homopolish` e `meyer2022critical` trazem a lista de autores truncada em "and others"; completar ou confirmar que o estilo aceita a forma abreviada.] Furthermore, while ConvNeXt proved more robust than ResNet50, isolating the specific components driving this advantage, such as Layer Normalization or large kernels, requires future ablation studies. Additionally, we observed high predictive confidence despite substantial performance degradation, particularly for the standard ResNet50; this behavior is consistent with overconfidence, although formal calibration was not evaluated.

Second, our interpretations of what the models learned remain associative. The strong local taxonomic consistency in the latent space does not necessarily mean the network reconstructed the true viral phylogeny, a hypothesis that requires validation against reference phylogenetic trees. Similarly, linking learned motifs to codon usage requires strictly controlling for broader genomic confounders, such as GC content and reading frames. Our sliding-window aggregation treats window-level predictions as independent, which is rarely true for overlapping windows. Future work should develop correlation-aware aggregation strategies to handle correlated errors and chimeric reads properly.

Also, the present evaluation was designed as a closed-set classification benchmark. All target taxonomic classes were represented during training, and dataset partitioning was performed at the sequence-fragment level. Therefore, the reported performance quantifies discrimination of held-out fragments among known taxa and does not directly measure generalization to previously unseen reference genomes or to species absent from the training label space.

Finally, several aspects of the experimental design limit the strength of the architectural comparison. Each configuration was trained once, so the variability attributable to random initialization and data ordering was not estimated. The training schedules and epoch budgets differed between architectures and initialization strategies, which confounds the attribution of performance differences to architecture alone. The KNN analysis used reference and query embeddings drawn from the same test set, so fragments from the same source genome may contribute to each other's neighborhoods. Label smoothing was used in the species-level training of the ConvNeXt models but not of the standard ResNet50, so differences in confidence behavior cannot be attributed to architecture alone. A controlled comparison with genomic foundation models would

require all models to be trained and evaluated under the same sequence partitions, preprocessing protocol, optimization budget, and corruption conditions.

4.2 Future Work

Future evaluation should prioritize more realistic long-read conditions, including empirical Oxford Nanopore reads and simulation protocols that incorporate substitutions, insertions, deletions, and homopolymer-associated errors. Formal calibration analysis using ECE, Brier Score, NLL, and post-hoc calibration methods would complement the confidence patterns observed here.

The biological interpretation of the latent space could be strengthened by directly comparing learned distances with reference phylogenetic distances. Similarly, motif analysis should be extended to annotated coding sequences with controls for reading frame, GC/GC3 content, dinucleotide composition, and other sequence-composition confounders.

Controlled architecture ablations would help determine which components of the evaluated ConvNeXt-inspired model contribute to corruption robustness. [**Nota v2:** [Sustentação] Para sustentar qualquer atribuição das diferenças à arquitetura, é necessário ao menos um par de modelos treinados sob condições idênticas: mesmo cronograma, mesmo número de épocas, mesmo *label smoothing* (com ou sem) e mesma inicialização. Sem esse controle, o artigo deve manter a formulação em termos de configurações, como feito na v2. Esse experimento também resolveria a questão da confiança: se o ResNet50 treinado com *label smoothing* passar a ter confiança decrescente, a diferença observada vem do treinamento, e não da arquitetura.] Taxon-dependent differences and recurrent prediction sinks also motivate ensemble or confidence-aware model-selection strategies.

Finally, sliding-window classification could be extended through learned aggregation mechanisms such as attention or multiple-instance learning. A controlled comparison with genomic foundation models using identical data partitions and corruption protocols would clarify the relative strengths of base-level convolutional and token-based genomic representations.

5 Conclusion

Base-level one-hot representations of viral genomic sequences can support high-cardinality taxonomic classification with 1D convolutional networks. ResNet50 and ConvNeXt exhibited distinct strengths: ResNet50 achieved superior performance on uncorrupted sequences, while ConvNeXt preserved predictive performance more effectively under sequence corruption and produced confidence estimates that better reflected degradation in classification quality, although this comparison is confounded by the use of label smoothing in the ConvNeXt training schedules.

The learned representations retained local taxonomic structure and sequence patterns with specific predictive effects. Local neighborhoods were strongly organized across species, genus, and order despite weak global cluster separation, while matched perturbation identified first-layer sequence patterns specifically associated with taxonomic predictions. ~~These validated motifs showed only weak and architecture-dependent associations with RSCU-derived nucleotide-triplet composition.~~ Exploratory analyses did not support a consistent

association between these motifs and RSCU-derived nucleotide-triplet composition. Sliding-window aggregation further shows that classifiers trained on fixed 512-bp fragments can be extended to variable-length sequences without modifying the underlying network architecture.

Evaluating genomic classification models solely by their clean-data accuracy provides an incomplete picture; a comprehensive assessment should also consider robustness, predictive uncertainty, latent-space organization, and interpretability. Within the evaluated closed-set setting, base-level one-hot nucleotide representations provide a practical approach to high-cardinality viral taxonomic classification. [Nota v2: [Contribuição] Reformular a Conclusão em torno da contribuição principal (o *trade-off* entre acurácia, robustez e confiança e o protocolo de avaliação), deixando explícito que a representação e as arquiteturas não são propostas como novidade.]

Supplementary information. If your article has accompanying supplementary file/s please state so here.

Authors reporting data from electrophoretic gels and blots should supply the full unprocessed scans for key as part of their Supplementary information. This may be requested by the editorial team/s if it is missing.

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Acknowledgements. Acknowledgements are not compulsory. Where included they should be brief. Grant or contribution numbers may be acknowledged.

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Declarations

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Appendix A Section title of first appendix

An appendix contains supplementary information that is not an essential part of the text itself but which may be helpful in providing a more comprehensive understanding of the research problem or it is information that is too cumbersome to be included in the body of the paper.

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